Pharmacovigilance Intelligence Platform: From Raw Adverse Event Reports to AI-Powered Safety Signals

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1. Dataset Overview

The source dataset used in this project consists of adverse event (AE) reports collected from a legacy pharmacovigilance safety system. Each record represents a single adverse event report submitted by healthcare professionals or consumers. The data includes patient demographics, drug information, event details, manufacturing information, and reporting metadata. The dataset was provided in CSV format and served as the primary input for the end-to-end data engineering pipeline.

1.1 Dataset Source:

Attribute	Value
Dataset Type	CSV
Domain	Pharmacovigilance
Source	Legacy Safety Database (Assignment Dataset)
Records	83
Columns	19

1.2 Dataset Schema:

Column	Description
REPORT_ID	Unique adverse event report identifier
REPORT_RECEIVED_DATE	Date the report was received
COUNTRY	Country Where the event was reported
REPORTER_TYPE	Healthcare Professional or Consumer
PATIENT_AGE	Patient age
PATIENT_SEX	Patient gender
DRUG_NAME	Suspected drug
DRUG_NDC_CODE	Drug Identification code

INDICATION	Therapeutic indication
ADVERSE_EVENT_TERM	Reported adverse event
EVENT_SERIOUSNESS	Serious or Non-Serious
OUTCOME	Outcome of the event
ONSET_DATE	Event onset date
BATCH_ID	Manufacturing batch
MANUFACTURING_SITE	Manufacturing location
DOSE_AMOUNT_MG	Administration dose
ROUTE_OF_ADMINISTRATION	Administration route
CONCOMITANT_MEDICATIONS	Other medications
CAUSALITY_ASSESSMENT	Initial causality assessment

1.3 **Raw Data Characteristics**

Initial data profiling revealed several quality issues that required cleansing before analytical use. The dataset contained duplicate adverse event reports, inconsistent text casing, missing values in patient age and dose amount, non-standardized date formats, and inconsistent categorical values. These issues were addressed during the Dataiku DSS data preparation stage.

1.4 **Initial Data Quality Assessment**

Issue	Description	Impact
Duplicate Records	Duplicate REPORT_ID Values	Duplicate reporting
Missing Patient Age	Null values	Incomplete demographics
Missing Dose	Null dose values	Reduced dosage analysis quality
Mixed Text Case	Drug/Country/Sex	Inconsistent grouping
Date Format	Multiple date representations	Date parsing issues
Missing Optional Fields	Concomitant medication	Incomplete clinical context

1.5 Raw Dataset Snapshot

Table Chart Pivot 83 rows 585ms

	REPORT_ID	REPORT_RECEIVED_DATE	COUNTRY	REPORTER_TYPE	PATIENT_AGE	PATIENT_SEX	DRUG_NAME	DRUG_NDC_CODE	INDICATION	ADVERSE_EVENT
	AE-202... 2.4% AE-202... 2.4% +79 more	09-19-2025 2.4% 2025-02-05 2.4% +67 more	Brazil 16.9% Germ... 13.3% +8 more	Consumer 21.7% physician 18.1% +5 more	Unknown 22.9% 24 2.4% +31 more	M 33.7% m 15.7% +3 more	Pulmoca... 14.5% Vasculin 14.5% +13 more	42858-7788-... 16.9% 12345-0456-... 15.7% +7 more	Deep Vein Thrombosis 16.9% COPD 15.7% +7 more	Hypotension Anaphylaxis +18 more
1	AE-2025-0001	2025-01-13	Canada	Physician	49	F	Neurolex	0069-5678-02	Generalized Anxiety Disorder	Headache
2	AE-2025-0002	2025-10-29	Japan	physician	18	null	Rheumatrix	0781-5566-11	Rheumatoid Arthritis	Peripheral edema
3	AE-2025-0003	2025-07-13	United Kingdom	Nurse	null	null	Neurolex	0069-5678-02	Generalized Anxiety Disorder	Dizziness
4	AE-2025-0004	2025-03-25	Germany	Nurse	63	null	Dermafex	16590-3344-01	Psoriasis	Dyspnoea
5	AE-2025-0005	2025-07-25	United Kingdom	physician	26	f	Dermafex	16590-3344-01	Psoriasis	Diarrhoea
6	AE-2025-0006	2025-07-05	Germany	physician	Unknown	M	Cardiozan	50090-1234-01	Hypertension	Pyrexia
7	AE-2025-0007	2025-05-17	USA	Other	null	m	Vasculin	42858-7788-02	Deep Vein Thrombosis	Neutropenia
8	AE-2025-0008	09-29-2025	Japan	physician	Unknown	M	VASCULIN	42858-7788-02	Deep Vein Thrombosis	Tachycardia
9	AE-2025-0009	2025-11-07	United Kingdom	Consumer	45	null	Vasculin	42858-7788-02	Deep Vein Thrombosis	Hypotension
10	AE-2025-0010	2025-12-29	Germany	Nurse	25	M	Neurolex	0069-5678-02	Generalized Anxiety Disorder	Thrombocytopenia
11	AE-2025-0011	2025-07-30	United Kingdom	NURSE	77	M	Cardiozan	50090-1234-01	Hypertension	Anaphylaxis
12	AE-2025-0012	2025-02-20	Brazil	physician	24	M	Vasculin	42858-7788-02	Deep Vein Thrombosis	Thrombocytopenia
13	AE-2025-0013	2025-12-05	USA	Nurse	37	U	Vasculin	42858-7788-02	Deep Vein Thrombosis	Headache
14	AE-2025-0014	2025-12-12	Australia	Physician	48	f	Neurolex	0069-5678-02	Generalized Anxiety Disorder	Pyrexia
15	AE-2025-0015	2025-02-07	Canada	Pharmacist	19	M	Dermafex	16590-3344-01	Psoriasis	Insomnia
16	AE-2025-0016	2025-06-03	Canada	physician	null	M	Vasculin	42858-7788-02	Deep Vein Thrombosis	Myalgia
17	AE-2025-0017	2025-05-22	Japan	physician	null	F	Cardiozan	50090-1234-01	Hypertension	Dyspnoea
18	AE-2025-0018	2025-01-22	Brazil	Other	57	U	Glucomend	00378-9012-10	Type 2 Diabetes	Dyspnoea
19	AE-2025-0019	2025-05-17	Brazil	physician	Unknown	M	Pulmocare-XR	12345-0456-05	COPD	Hypotension
20	AE-2025-0020	2025-05-23	Canada	Pharmacist	Unknown	null	Neurolex	0069-5678-02	Generalized Anxiety Disorder	Rash
21	AE-2025-0021	09-19-2025	USA	Nurse	67	U	Immunova	68180-9900-04	Ulcerative Colitis	Dyspnoea
22	AE-2025-0022	2025-12-08	India	physician	Unknown	null	Oncozel	60505-1122-03	Non-small Cell Lung Cancer	Diarrhoea

2. Solution Architecture

2.1 Architecture Overview

Description:

The Pharmacovigilance Intelligence Platform was designed as a cloud-native, end-to-end data engineering solution for processing and analyzing adverse event (AE) reports. The architecture integrates cloud storage, ETL, data warehousing, machine learning, and business intelligence into a unified analytical platform.

The solution begins with raw adverse event reports stored in Amazon S3. Dataiku DSS is then used to profile, clean, and standardize the data before loading it into Snowflake. Within Snowflake, the data is organized into RAW, STAGING, and CURATED layers using a medallion architecture. A dimensional star schema is created in the CURATED layer to support analytical queries and machine learning.

Snowflake Cortex ML is then used to detect unusual spikes in adverse event reporting by analyzing aggregated monthly adverse event counts. Finally, the curated data and AI-generated anomaly results are visualized in an interactive Power BI dashboard to support pharmacovigilance safety review and decision-making.

2.2 **Architecture Objectives**

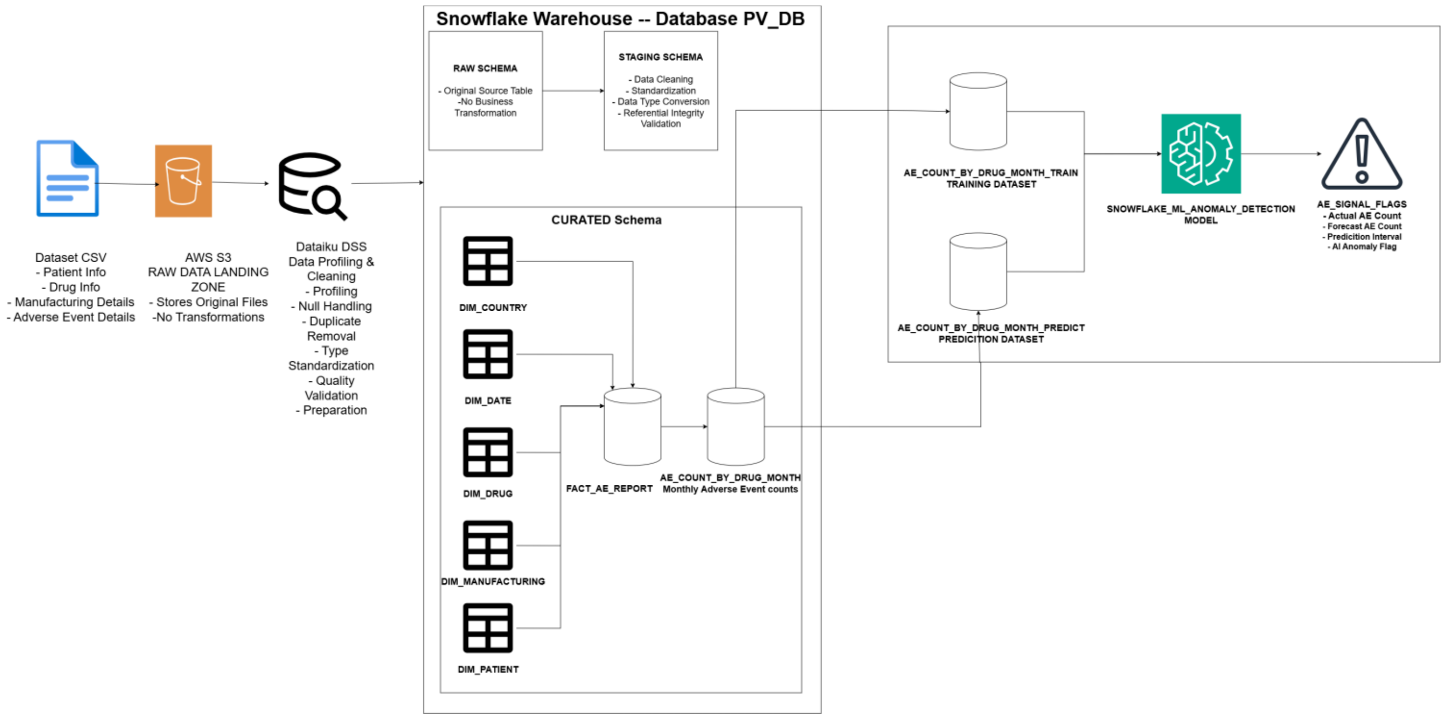
The architecture was designed to achieve the following objectives:

- Store raw adverse event reports in a secure cloud storage environment.
- Implement an auditable ETL process using Dataiku DSS.
- Build a production-ready dimensional model in Snowflake.
- Generate analytical datasets optimized for reporting and machine learning.
- Detect statistical anomalies using Snowflake Cortex ML.
- Deliver interactive business intelligence dashboards for safety monitoring.

2.3 **Architecture Components**

Component	Technology	Purpose
Data Source	CSV Dataset	Source of adverse event reports
Cloud Storage	AWS S3	Stores raw input files
ETL platform	Dataiku DSS	Profiling, cleansing and standardization
Data Warehouse	Snowflake	Central analytical database
Data Model	Star Schema	Optimized analytical model
Machine Learning	Snowflake Cortex ML	Statistical anomaly detection
Reporting	Power BI	Interactive dashboard and visualization

2.4 **End-to-End Architecture**



2.5 **End-to-End Data Flow**

Step 1 – Raw Data Acquisition

Adverse event reports are received as CSV extracts from the legacy pharmacovigilance safety database.

Step 2 – Cloud Storage

The CSV files are uploaded to Amazon S3 using a structured folder hierarchy to separate raw and processed data.

Step 3 – Data Profiling and Cleansing

Dataiku DSS profiles the source dataset, identifies quality issues, removes duplicates, standardizes categorical values, normalizes dates, and prepares the dataset for analytical processing.

Step 4 – Snowflake Data Warehouse

The cleaned dataset is loaded into Snowflake, where it passes through the RAW, STAGING, and CURATED layers.

Step 5 – Dimensional Modeling

The curated layer implements a star schema consisting of one fact table and multiple dimension tables to support efficient analytical queries.

Step 6 – Aggregation Layer

The fact table is aggregated into monthly adverse event counts by drug to produce a time-series dataset suitable for machine learning.

Step 7 – Snowflake Cortex ML

The aggregated data is divided into training and prediction datasets. Snowflake Cortex ML trains an anomaly detection model and identifies statistically unusual reporting patterns.

Step 8 – Business Intelligence

The curated data model and anomaly detection results are consumed by Power BI to create an interactive pharmacovigilance dashboard for business users.

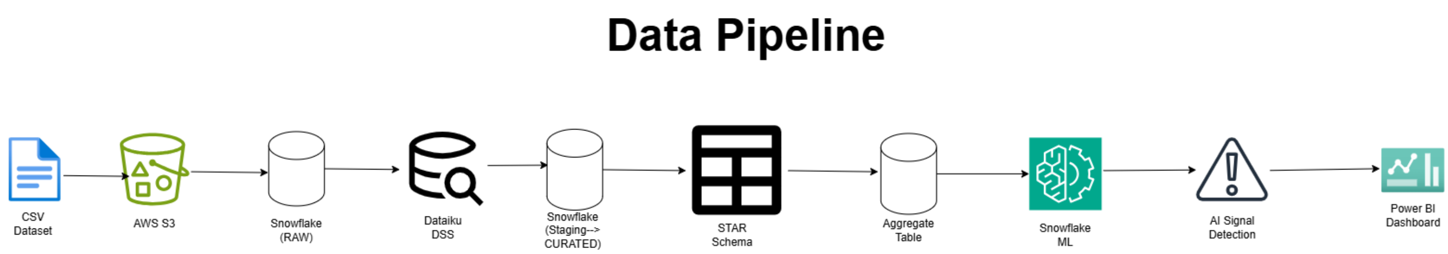
2.6 **Architecture Benefits**

The proposed architecture provides several advantages:

- **Scalability:** Cloud-native components support increasing volumes of adverse event data.
- **Data Quality:** Automated profiling and validation improve data reliability.
- **Performance:** The star schema enables efficient analytical querying.
- **Governance:** Layered architecture improves traceability and maintainability.

- **AI Integration:** Snowflake Cortex ML enables statistical signal detection without external ML platforms.
- **Business Insights:** Power BI provides intuitive dashboards for safety reviewers and decision-makers.

2.7 **Data Pipeline**



3. **AWS Cloud Storage and Data Ingestion**

3.1 Overview

Amazon Web Services (AWS) was used as the cloud storage platform for the Pharmacovigilance Intelligence Platform. The adverse event dataset was uploaded to an Amazon S3 bucket, which served as the landing zone for raw data before ingestion into Snowflake.

Using Amazon S3 decouples data storage from data processing and provides a scalable, durable, and secure repository for source files.

3.2 **Objectives**

The AWS implementation was designed to achieve the following objectives:

- Store raw adverse event reports in cloud storage.
- Organize source data using structured folder prefixes.
- Enable secure integration between Amazon S3 and Snowflake.
- Support automated data ingestion through Snowflake External Stages.

3.3 **AWS Services Used**

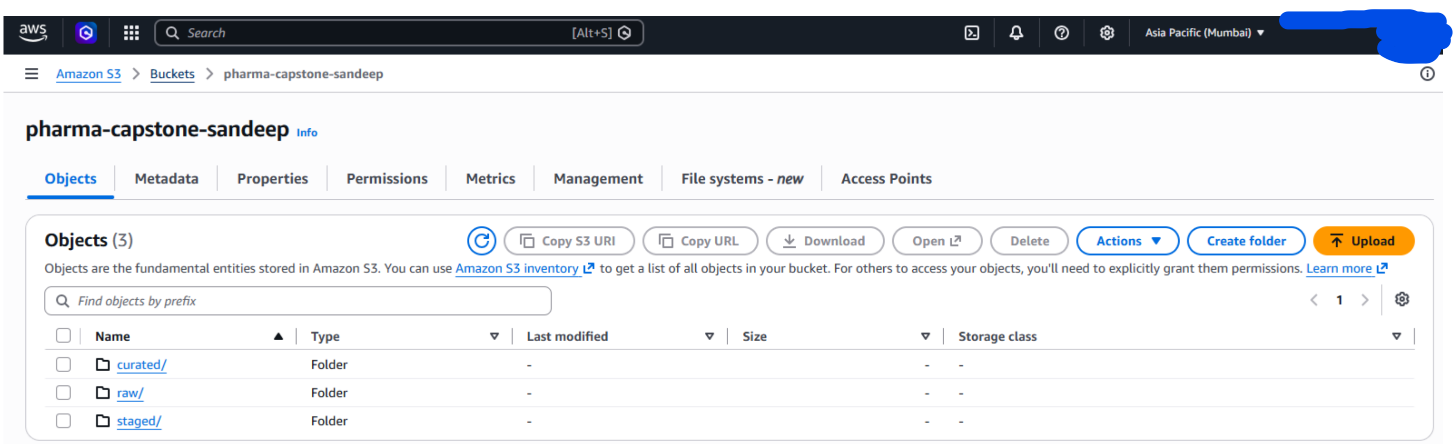
AWS Service	Purpose
Amazon S3	Cloud Object storage for raw dataset
AWS IAM	Secure authentication and authorization between AWS and Snowflake

3.4 **Amazon S3 Bucket Structure**

A dedicated Amazon S3 bucket was created to store the pharmacovigilance dataset.

Bucket Name: pharma-capstone-sandeep

Folder Structure:



Purpose of Each Folder:

Folder	Purpose
Raw/	Stores Original source CSV files exactly as received
Staging/	Reserved for intermediate processed datasets
Curated/	Reserved for final business-ready datasets

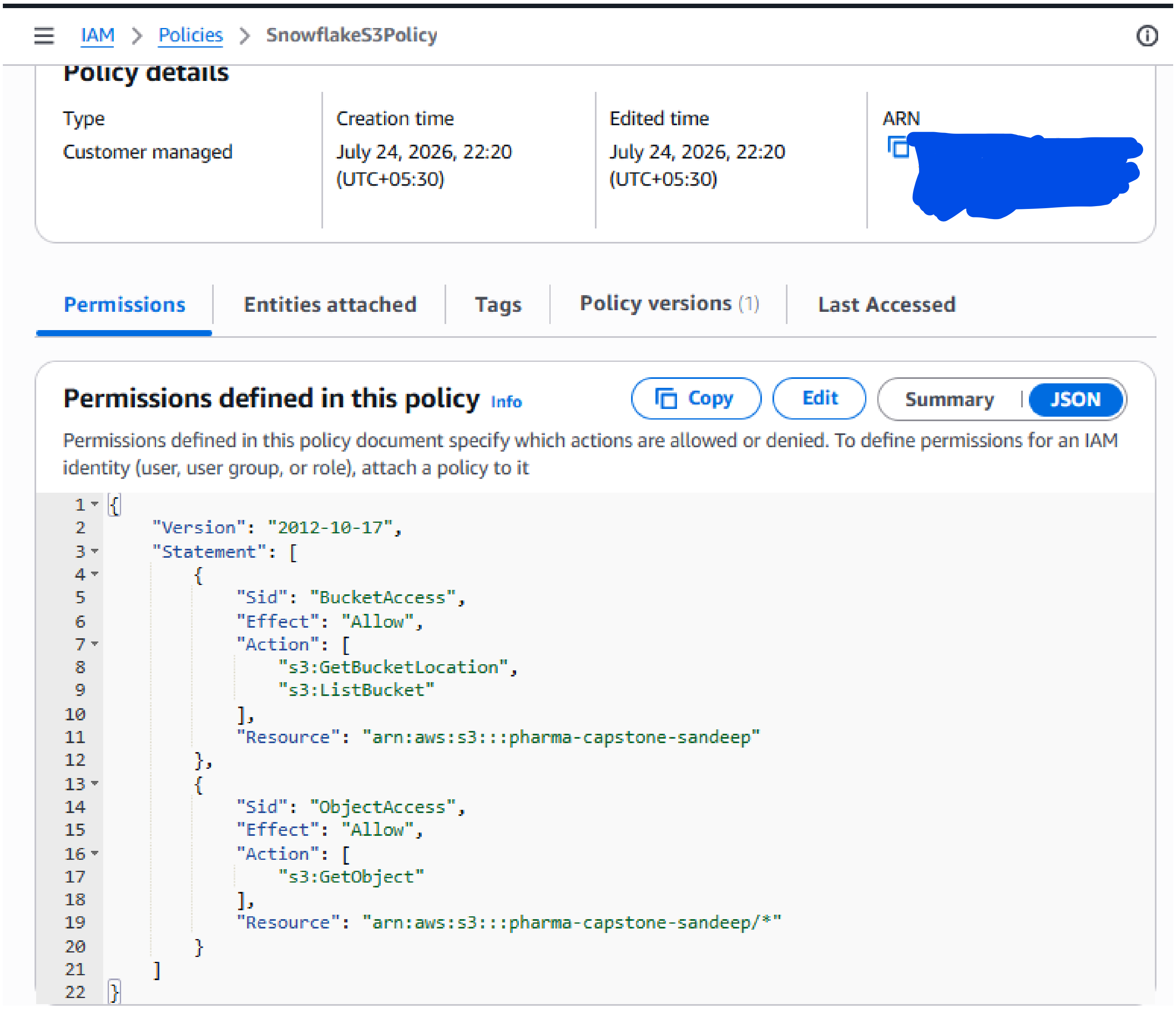
3.5 **IAM Configuration**

To allow Snowflake to securely access files stored in Amazon S3, an AWS Identity and Access Management (IAM) role was created.

IAM Role: snowflakes3role

The IAM role grants Snowflake read access to the Amazon S3 bucket without exposing AWS access keys.

The trust relationship between AWS and Snowflake enables secure cross-account authentication through a Storage Integration.



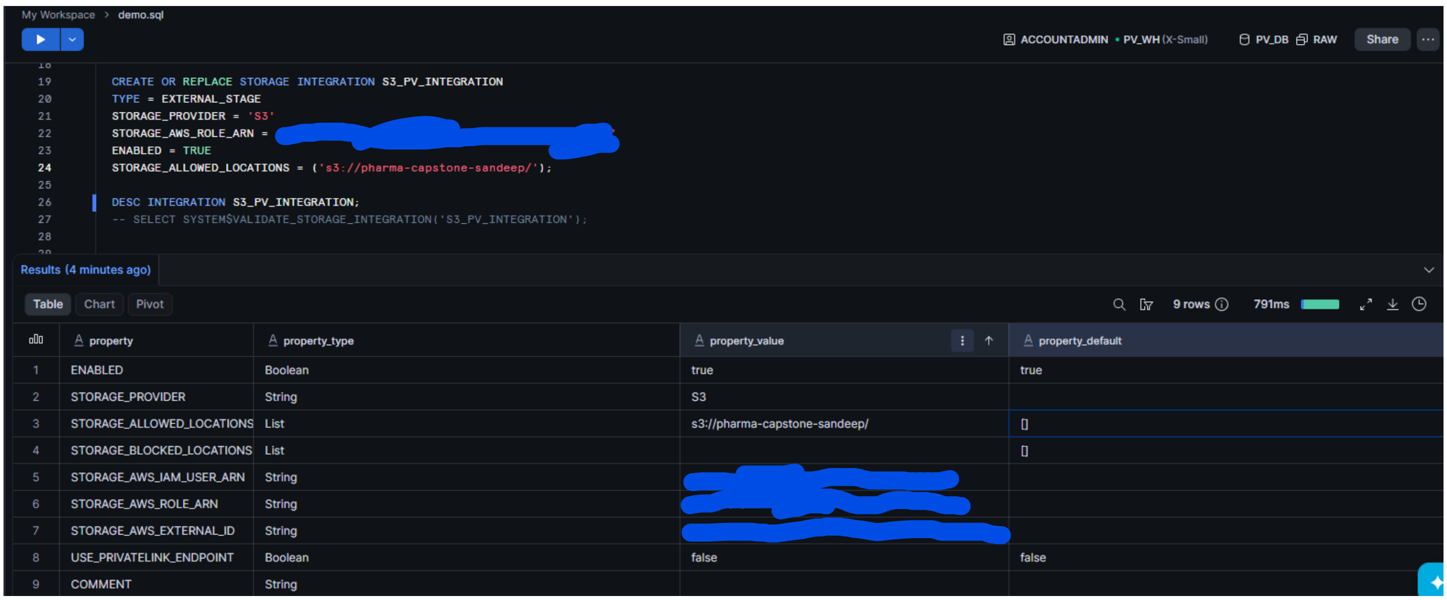
3.6 **Snowflake Storage Integration**

A Snowflake Storage Integration was configured to establish a secure connection with Amazon S3.

Storage Integration: S3_PV_INTEGRATION

Configuration Summary:

- Storage Provider: Amazon S3
- Authentication: AWS IAM Role
- Allowed Location: s3://pharma-capstone-sandeep/



3.7 **External Stage**

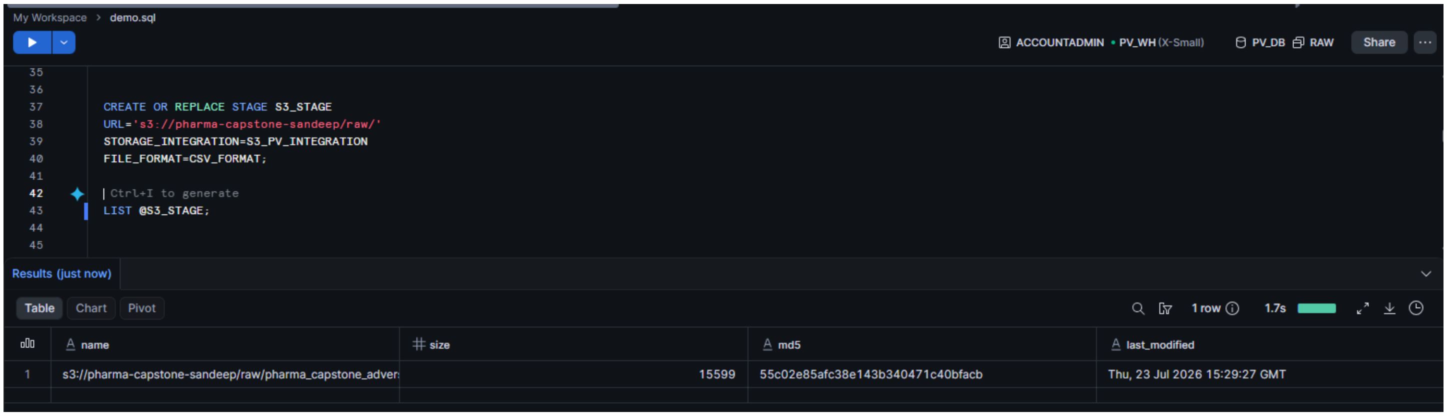
A Snowflake External Stage was created to reference the raw data stored in Amazon S3.

External Stage: S3_STAGE

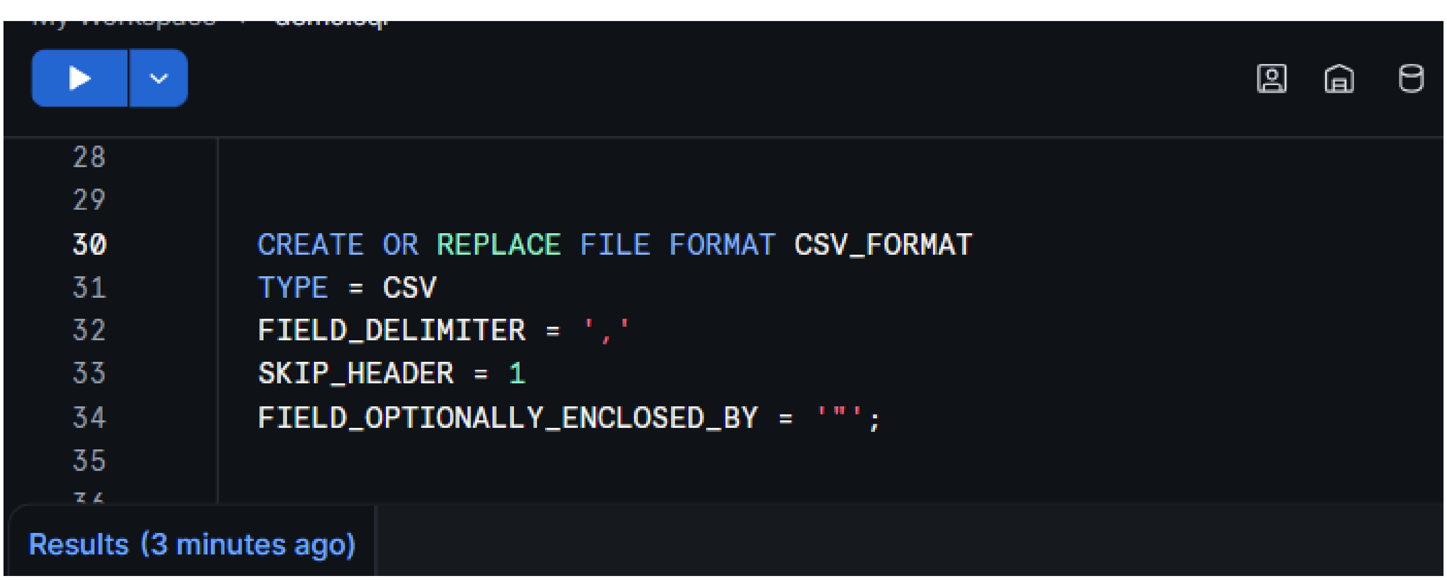
Location: s3://pharma-capstone-sandeep/raw/

Purpose

- Access files stored in Amazon S3.
- Read CSV files without manually downloading them.
- Serve as the source for Snowflake COPY INTO operations.

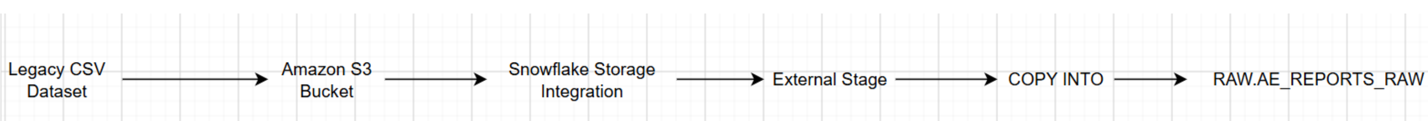


3.8 <u>**File Format**</u>



3.9 <u>**Data Ingestion Process**</u>

The raw adverse event dataset was ingested into the Snowflake RAW layer using the following process:



```sql
COPY INTO RAW.AE_REPORTS_RAW
FROM @CURATED.S3_STAGE
FILE_FORMAT = (FORMAT_NAME = CURATED.CSV_FORMAT);


SELECT COUNT(*) FROM RAW.AE_REPORTS_RAW;
```

9 minutes ago)

Chart Pivot 1 row 118ms

COUNT(*)
83

Success Criteria

1.RAW Table raw count matches the source file:

My Workspace > demo.sql

Share

```sql
SELECT
(
    SELECT COUNT(*)
    FROM PV_DB.RAW.AE_REPORTS_RAW
) AS ROW_COUNT,

(
    SELECT COUNT(*)
    FROM PV_DB.INFORMATION_SCHEMA.COLUMNS
    WHERE TABLE_SCHEMA = 'RAW'
      AND TABLE_NAME = 'AE_REPORTS_RAW'
) AS COLUMN_COUNT;
```

Results (just now)

Table Chart Pivot 1 row 1.1s

	ROW_COUNT	COLUMN_COUNT
1	83	19

2. At least 6 distinct data quality issues documented

Issue	Description	Impact
Duplicate Records	Duplicate REPORT_ID Values	Duplicate reporting
Missing Patient Age	Null values	Incomplete demographics
Missing Dose	Null dose values	Reduced dosage analysis quality
Mixed Text Case	Drug/Country/Sex	Inconsistent grouping
Date Format	Multiple date representations	Date parsing issues
Missing Optional Fields	Concomitant medication	Incomplete clinical context

4. **Dataiku DSS: Data Profiling, Cleansing and Standardization**

Firstly, Snowflake and Dataiku are connected to get the raw data from snowflake.

4.1 **Overview**

Dataiku DSS was used as the Extract, Transform, and Load (ETL) platform to profile, cleanse, and standardize the raw adverse event dataset before loading it into the Snowflake data warehouse. It provided a visual workflow for data preparation while maintaining an auditable sequence of transformations.

The objective of this stage was to improve data quality, remove inconsistencies, and ensure that only validated records were loaded into the analytical data warehouse.

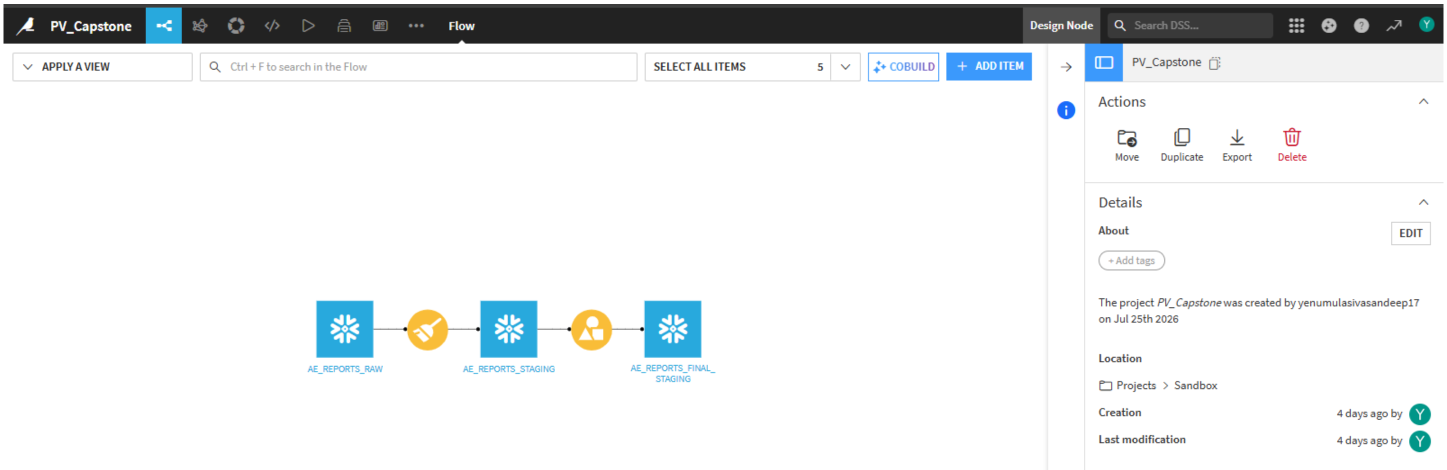
4.2 **Objectives**

The Dataiku ETL workflow was implemented to achieve the following objectives:

- Profile the raw adverse event dataset.
- Identify data quality issues.
- Remove duplicate records.
- Standardize categorical values.
- Normalize date formats.
- Handle missing values.

- Validate data quality before loading into Snowflake.
- Load the cleaned dataset into the STAGING schema.

4.3 **Dataiku Workflow**



The Dataiku workflow begins with the raw adverse event dataset imported from Snowflake Raw Schema. A sequence of preparation recipes is applied to profile the data, clean invalid records, standardize values, and perform quality validation. The final cleaned dataset is exported to the Snowflake STAGING schema for dimensional modeling.

4.4 **Data Profiling**

The first activity performed in Dataiku DSS was data profiling.

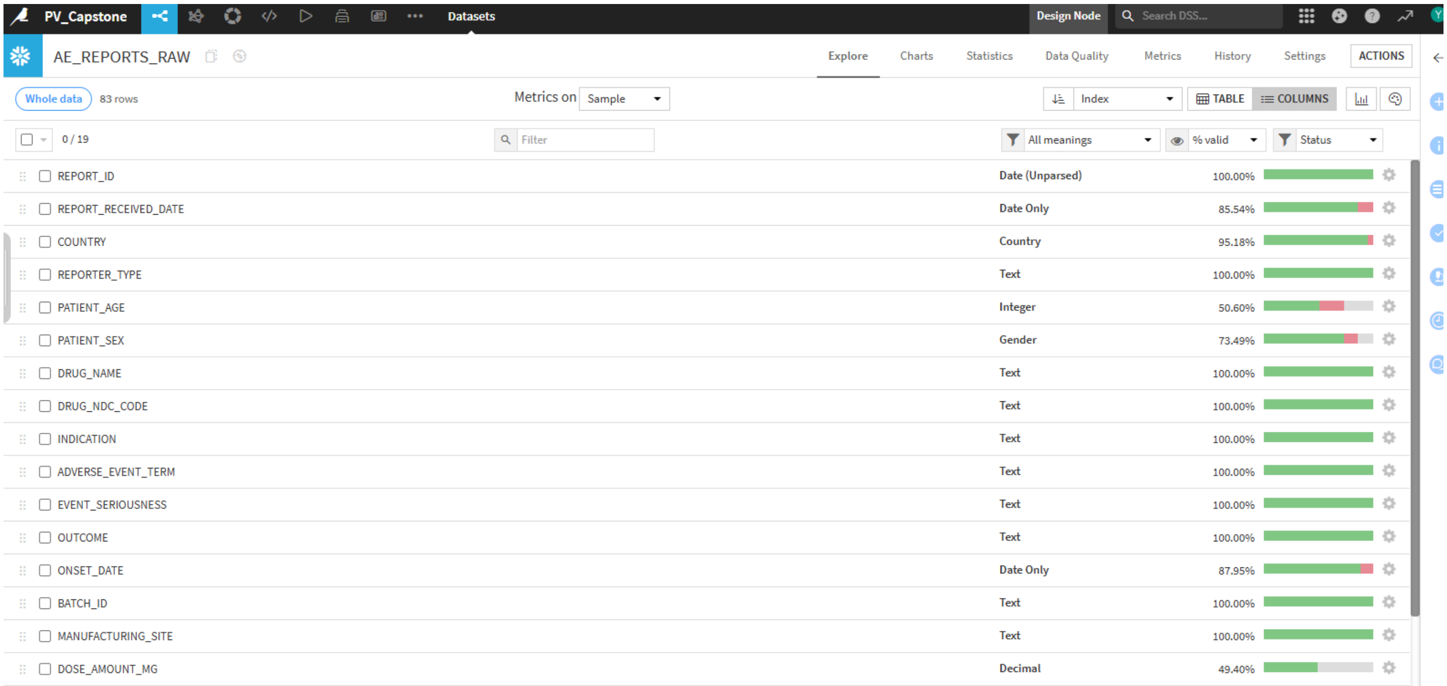
Profiling provides a statistical summary of the dataset and helps identify quality issues before transformation.

The following aspects were analyzed:

- Total number of records
- Total number of columns
- Null values
- Duplicate records
- Data types
- Missing values

- Value distributions
- Invalid formats

This step established the baseline quality of the incoming dataset.



4.5 **Data Quality Issues Identified**

During profiling, several data quality issues were identified.

Issue	Description	Resolution
Duplicate Report IDs	Multiple records with the same REPORT_ID	Removed duplicate records
Missing Patient Age	Null values in PATIENT_AGE	Retained as NULL for later analysis
Missing Dose Amount	Null values in DOSE_AMOUNT_MG	Retained as NULL
Inconsistent Text Case	Mixed uppercase/lowercase values	Standardized text formatting
Non-standard Date Formats	Dates stored as in Wrong Format	Converted to DATE format
Inconsistent Country Names	Text formatting inconsistencies	Standardized using proper case

4.6 **Data Cleansing**

Duplicate Removal

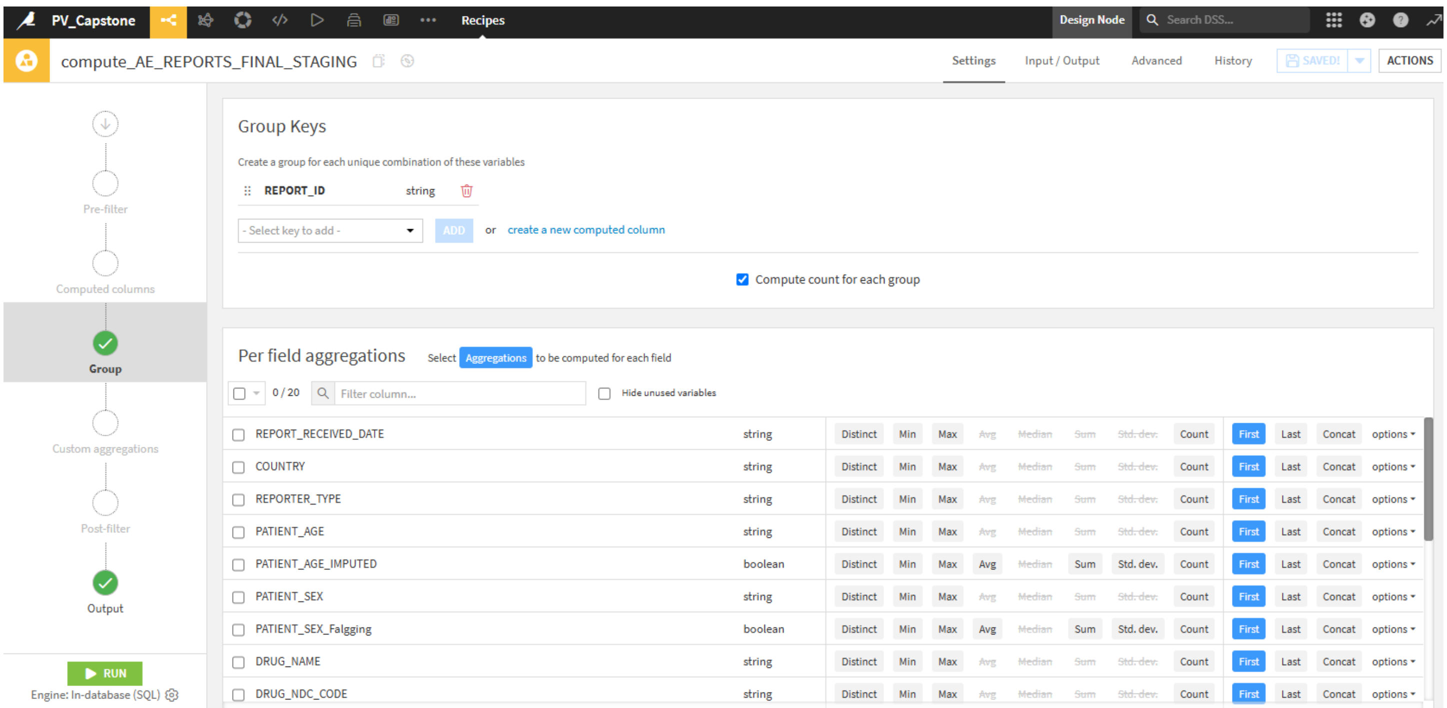
Duplicate adverse event reports were identified using the REPORT_ID column.

Business Rule:

If multiple records share the same REPORT_ID, retain the first occurrence and remove subsequent duplicates.

Outcome:

- Duplicate reports removed
- Unique REPORT_ID values maintained



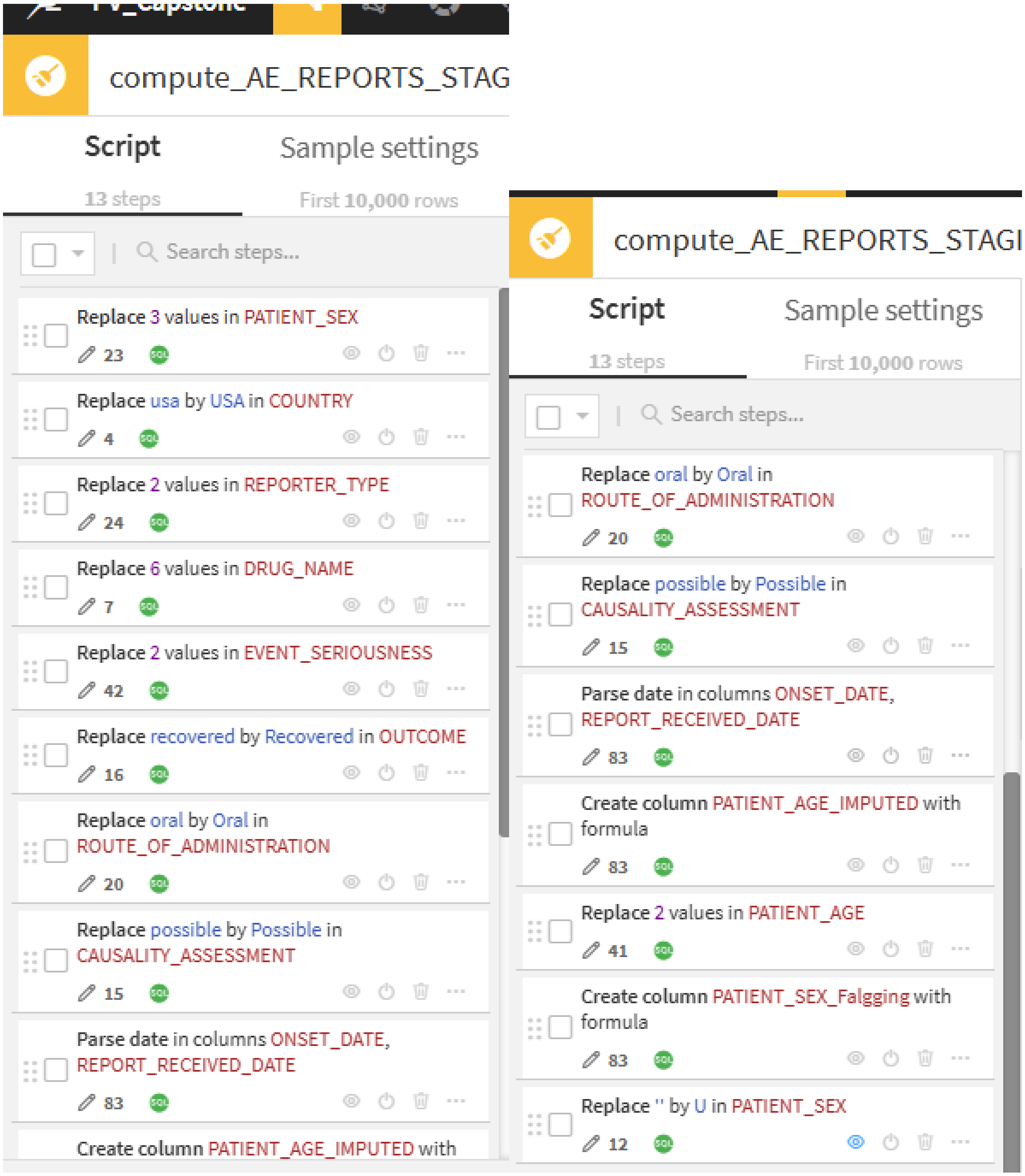
Missing Value Handling

Missing values were analyzed for all columns.

Handling strategy:

Column	Strategy
PATIENT_AGE	Retained with Median Value
DOSE_AMOUNT_MG	Retained with Null
PATIENT_SEX	Retained with Unknown(“U”)
Optional Clinical Fields	Retained as Null

The missing values were preserved because imputing clinical information could introduce incorrect assumptions into the pharmacovigilance analysis.



Text Standardization

Categorical columns were standardized to ensure consistency.

Examples include:

- Drug Name
- Country
- Patient Sex
- Reporter Type
- Manufacturing Site

Standardization eliminated inconsistencies caused by mixed text casing.

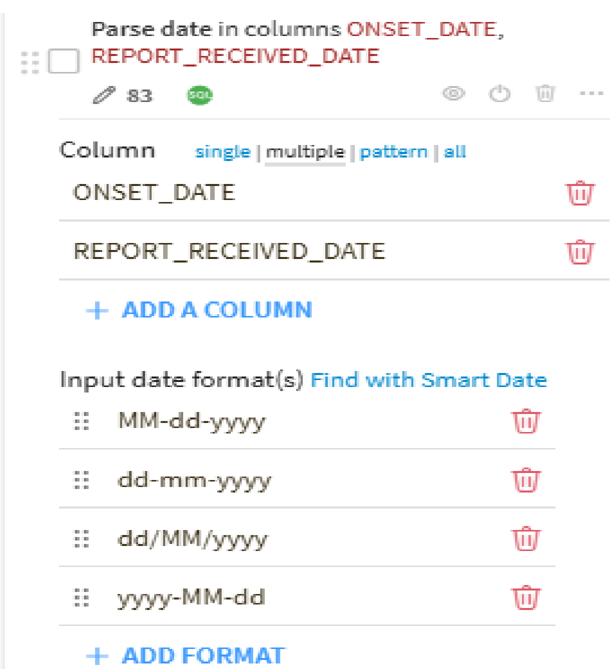
Date Normalization

Date fields were converted into the standard ISO format (YYYY-MM-DD).

The following columns were normalized:

- REPORT_RECEIVED_DATE
- ONSET_DATE

This ensured compatibility with Snowflake DATE data types.



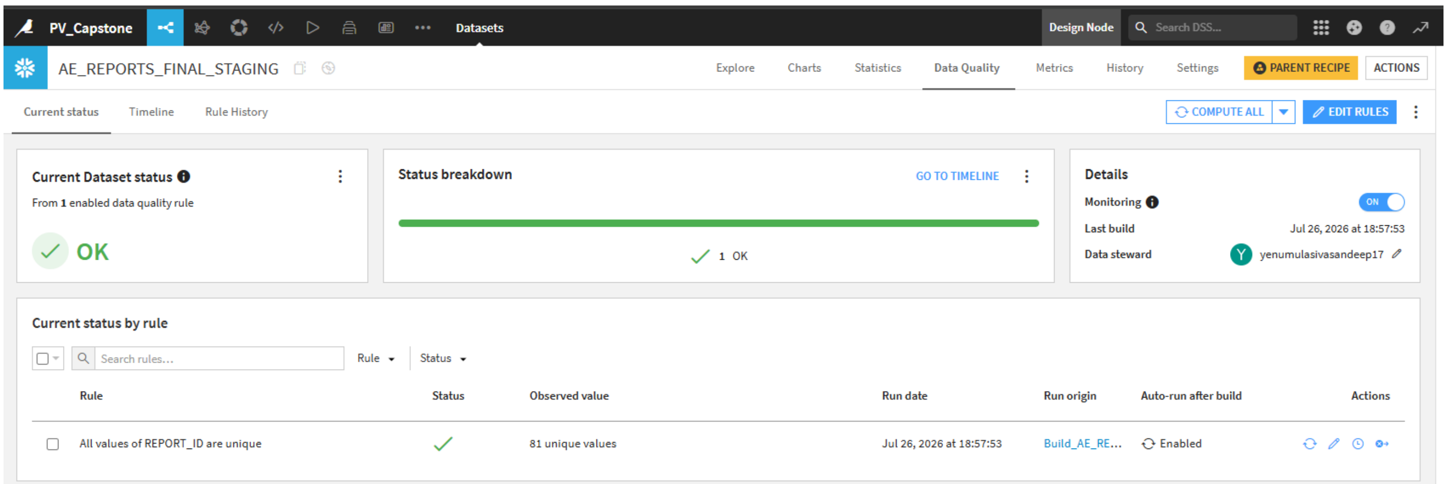
4.7 **Data Quality Checkpoint**

A Dataiku quality checkpoint was configured to validate the cleaned dataset before export.

The checkpoint verifies that:

- Duplicate REPORT_ID values do not exist.

If validation fails, the workflow stops until the issues are resolved.

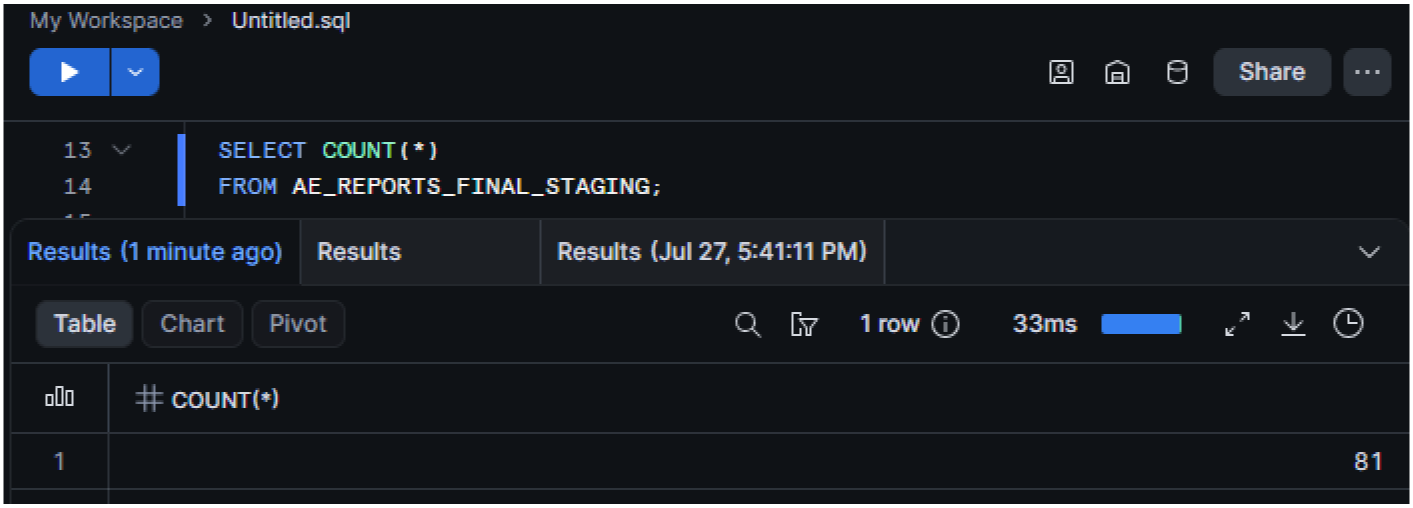


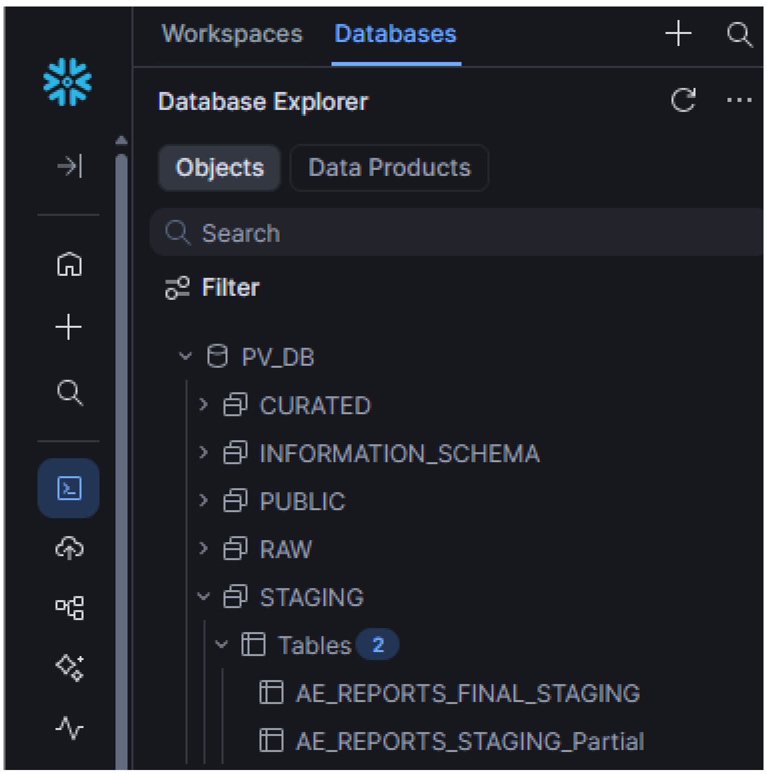
4.8 **<u>Export to Snowflake</u>**

After successful validation, the cleaned dataset was exported to the Snowflake STAGING schema.

Destination: PV_DB.STAGING.AE_REPORTS_FINAL_STAGING

The exported dataset serves as the source for dimension table population and fact table loading.


My Workspace > Untitled.sql
Share
13 SELECT COUNT(*)
14 FROM AE_REPORTS_FINAL_STAGING;
Results (1 minute ago)
Results
Results (Jul 27, 5:41:11 PM)
Table
Chart
Pivot
1 row
33ms
COUNT(*)
1
81


Workspaces
Databases
Database Explorer
Objects
Data Products
Search
Filter
PV_DB
CURATED
INFORMATION_SCHEMA
PUBLIC
RAW
STAGING
Tables 2
AE_REPORTS_FINAL_STAGING
AE_REPORTS_STAGING_Partial

4.9 DQ RuleBook

DQ stands for **Data Quality**.

This Tells **all the quality rules** applied during the ETL process, mainly in Dataiku DSS.

Rule ID	Quality Rule	Action Taken
DQ-001	Duplicate REPORT_ID	Remove duplicates
DQ-002	REPORT_RECEIVED_DATE must be valid	Convert to DATE
DQ-003	ONSET_DATE must be valid	Convert to DATE
DQ-004	DRUG_NAME must be standardized	Standardize casing
DQ-005	COUNTRY must be standardized	Standardize casing
DQ-006	PATIENT_SEX values standardized	Added U (Imputation)
DQ-007	Missing PATIENT_AGE	Retains Median (Imputation)
DQ-008	Missing DOSE_AMOUNT_MG	Retain NULL (no imputation)
DQ-009	REPORT_ID must be unique	Quality checkpoint
DQ-010	Data loaded to STAGING only after validation	Workflow checkpoint

4.10 Data Quality Rule Descriptions

DQ-001 – Duplicate REPORT_ID

Each adverse event report must have a unique `REPORT_ID`. Duplicate records were identified using the `REPORT_ID` field and removed during the Dataiku DSS data preparation process. The first valid occurrence of each report was retained while duplicate entries were discarded. This ensured that every adverse event report was represented only once in the staging dataset.

DQ-002 – REPORT_RECEIVED_DATE Validation

The `REPORT_RECEIVED_DATE` field must contain valid calendar dates in the standard `YYYY-MM-DD` format. During data preparation, all values were validated and converted to the Snowflake DATE data type. This ensured consistency in date representation and enabled accurate time-based analysis.

DQ-003 – ONSET_DATE Validation

The `ONSET_DATE` field represents the date on which the adverse event occurred and must contain valid date values. The column was validated and converted to the standard DATE format. This transformation ensured consistent storage and improved compatibility with analytical queries and machine learning models.

DQ-004 – DRUG_NAME Standardization

Drug names contained inconsistent text casing due to variations in data entry. All drug names were standardized using a consistent naming convention to eliminate duplicate representations of the same drug. This improved grouping, aggregation, and reporting accuracy across the analytical model.

DQ-005 – COUNTRY Standardization

Country names were standardized to ensure consistent formatting throughout the dataset. Variations in capitalization were corrected so that each country appeared in a single standardized format. This improved the accuracy of country-level reporting and geographical analysis.

DQ-006 – PATIENT_SEX Standardization

Missing or undefined values in the `PATIENT_SEX` column were replaced with the value **'U' (Unknown)**. This ensured that every record contained a valid categorical value while preserving the fact that the original information was unavailable. The standardization improved data consistency without introducing incorrect demographic information.

DQ-007 – Missing PATIENT_AGE

Missing values in the `PATIENT_AGE` column were handled using **median imputation**. The median age of the available records was calculated and used to populate missing values. This approach minimized the impact of missing data while preserving the overall age distribution and reducing the influence of extreme values.

DQ-008 – Missing DOSE_AMOUNT_MG

Missing values in the `DOSE_AMOUNT_MG` column were intentionally retained as `NULL`. Since dosage information is clinical in nature, no imputation was performed to avoid introducing artificial or misleading values. Retaining NULL values preserves the integrity of the original clinical data.

DQ-009 – REPORT_ID Quality Checkpoint

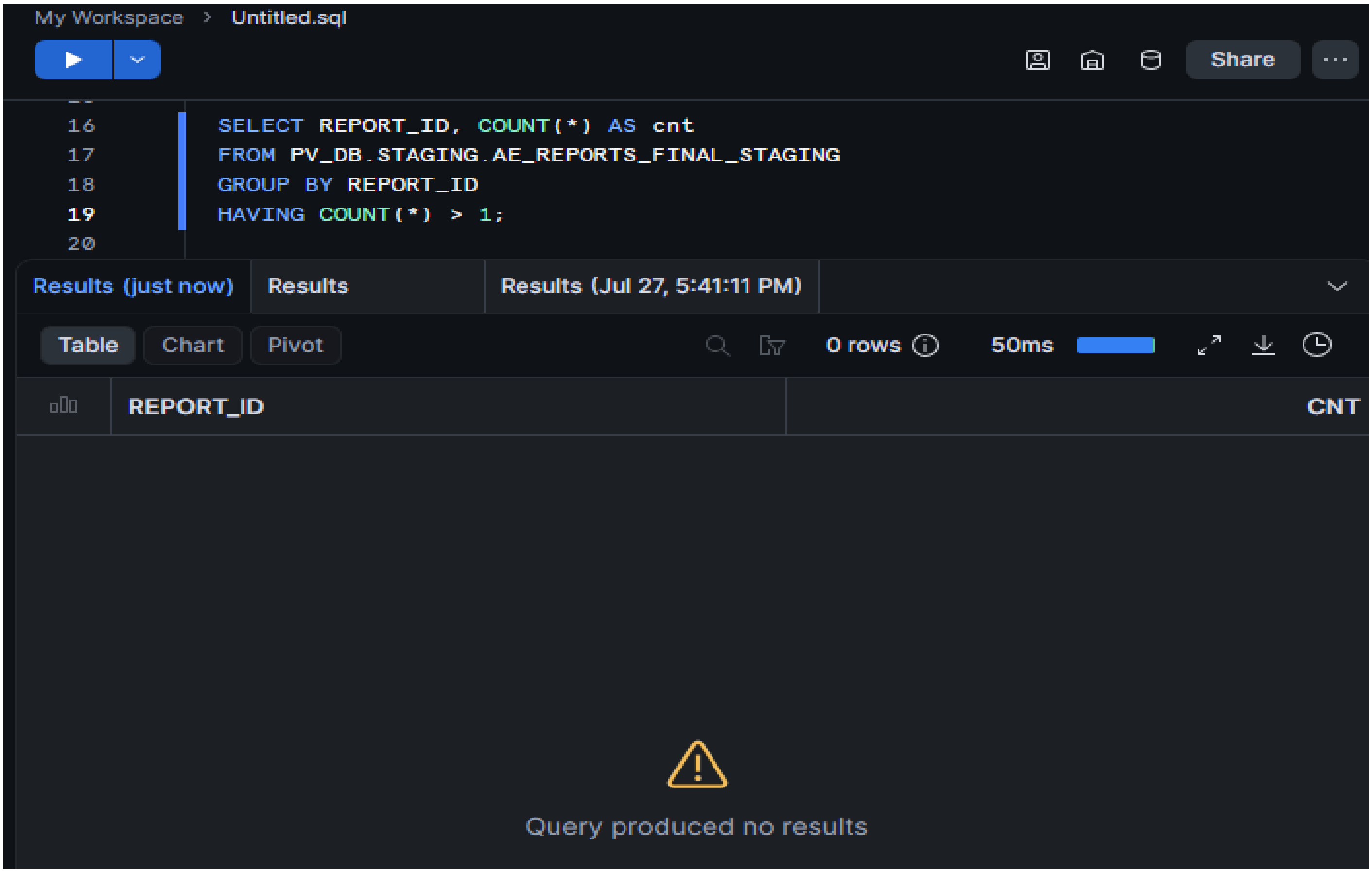
A Dataiku Data Quality Checkpoint was configured to verify that all `REPORT_ID` values were unique before the dataset was exported to Snowflake. If duplicate report identifiers were detected, the workflow would fail validation and prevent invalid data from progressing further in the ETL pipeline.

DQ-010 – Workflow Validation Checkpoint

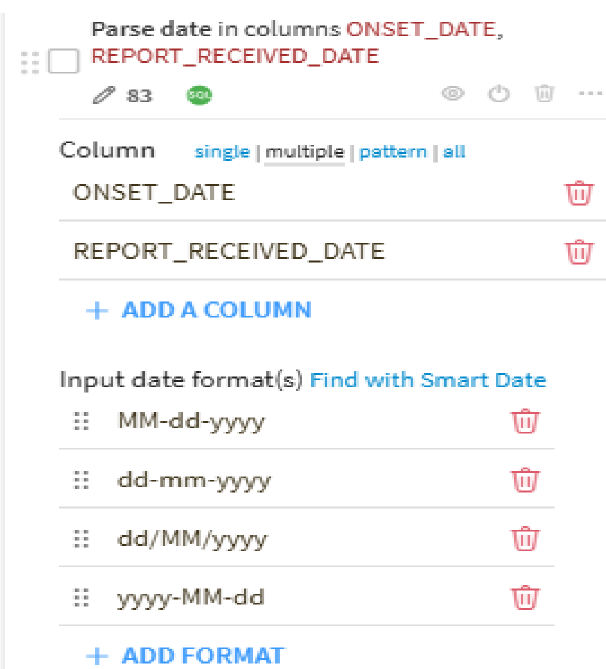
A final workflow validation checkpoint was implemented to ensure that only datasets satisfying all predefined data quality rules were loaded into the Snowflake **STAGING** schema. The ETL workflow proceeded to the next stage only after all quality checks were successfully completed, ensuring that downstream dimensional modeling, machine learning, and reporting operated on validated and reliable data.

Success Criteria

Zero Duplicate report_ids in staging



All dates valid as DATE type



5. <u>Snowflake Data Warehouse</u>

5.1 Overview

Snowflake was used as the enterprise cloud data warehouse for storing, transforming, and analyzing the adverse event (AE) data. Following data cleansing in Dataiku DSS, the processed dataset was loaded into Snowflake, where it was organized using a layered architecture consisting of RAW, STAGING, and CURATED schemas.

This architecture separates data ingestion, transformation, and business-ready analytical datasets, making the solution scalable, maintainable, and suitable for production-style data engineering workflows.

5.2 Objectives

The Snowflake implementation was designed to achieve the following objectives:

- Store cleaned adverse event data in a centralized cloud data warehouse.
- Implement a layered architecture for better data governance.
- Separate raw, staging, and curated datasets.
- Design a production-ready dimensional model.
- Support analytical reporting and machine learning.
- Provide a governed data source for Power BI dashboards.

5.3 Snowflake Environment

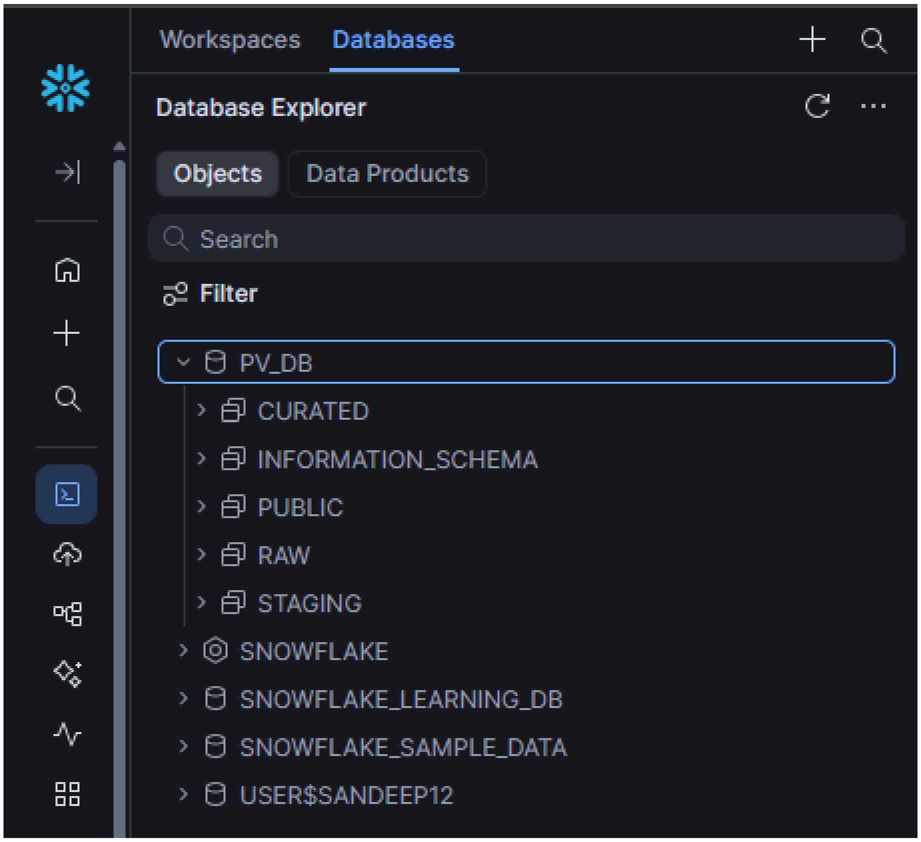
The following Snowflake objects were created for this project.

Object	Name	Purpose
Warehouse	**PV_WH**	Compute resources for SQL execution
Database	**PV_DB**	Central pharmacovigilance database
Schema	**RAW**	Raw ingested data
Schema	**STAGING**	Cleaned and standardized data

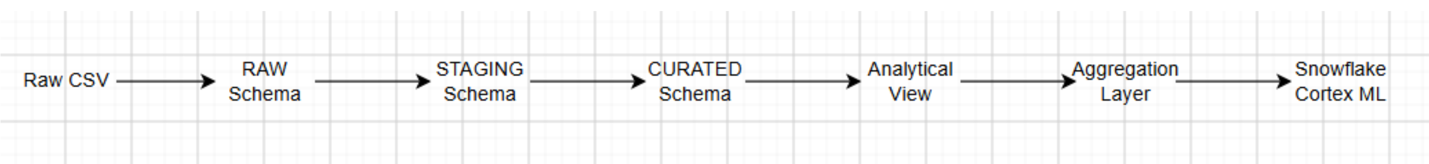
Schema	**CURATED**	Business-ready dimensional model

5.4 <u>Layered Architecture</u>

The Snowflake environment follows a Medallion-style layered architecture.



5.5 <u>Data Flow Through Snowflake</u>



5.6 <u>Benefits of Layered Architecture</u>

The layered architecture provides several advantages.

Data Isolation:

Each stage of data processing is isolated, reducing the risk of accidental modification to source data.

Improved Data Quality:

Data cleansing occurs only within the STAGING layer, ensuring that downstream analytical models receive standardized data.

Easier Maintenance:

Separating ingestion, transformation, and analytics simplifies debugging and future enhancements.

Better Performance:

Business users query only the CURATED layer, which contains optimized tables and views.

Data Lineage:

Every record can be traced from the curated model back to its raw source.

6.STAGING Schema

6.1 Overview

The **STAGING** schema serves as the intermediate processing layer between the RAW and CURATED schemas. After the raw adverse event dataset was profiled and cleansed in Dataiku DSS, the cleaned output was loaded into the STAGING schema. This layer contains standardized, validated, and transformation-ready data that acts as the source for dimensional modeling.

Unlike the RAW schema, which preserves the original source data, the STAGING schema contains processed records that have undergone data quality improvements while maintaining the original business meaning.

6.2 **Objectives**

The STAGING schema was implemented to achieve the following objectives:

- Store the cleaned output from Dataiku DSS.
- Maintain standardized and validated data.
- Prepare the dataset for dimensional modeling.
- Separate transformation logic from analytical models.
- Provide a reliable source for loading dimension and fact tables.

Table Chart Pivot 81 rows 59ms

	REPORT_ID	REPORT_RECEIVED_DATE_first	COUNTRY_first	REPORTER_TYPE_first	PATIENT_AGE_first	PATIENT_AGE_IMPUTED_first	PATIENT_SEX_first	PATIENT_SEX_Falgging_first	DRU
	AE-202... 1.2%	2025-02-05 2.5%	Brazil 17.3%	Physician 34.6%	54.5 50.6%		M 49.4%		Vasculi
	AE-202... 1.2%	2025-02-22 2.5%	Germany 13.6%	Consumer 22.2%	24 2.5%	true 50.6%	U 25.9%	false 85.2%	Pulmoc
	+79 more	+65 more	+7 more	+3 more	+31 more	false 49.4%	+1 more	true 14.8%	+7 mor
1	AE-2025-0002	2025-10-29	Japan	Physician	18	FALSE	U	TRUE	Rheuma
2	AE-2025-0009	2025-11-07	United Kingdom	Consumer	45	FALSE	U	TRUE	Vasculi
3	AE-2025-0011	2025-07-30	United Kingdom	Nurse	77	FALSE	M	FALSE	Cardioz
4	AE-2025-0014	2025-12-12	Australia	Physician	48	FALSE	F	FALSE	Neurole
5	AE-2025-0025	2025-12-23	Germany	Consumer	84	FALSE	F	FALSE	Glucom
6	AE-2025-0027	2025-02-02	USA	Physician	54.5	TRUE	F	FALSE	Pulmoc
7	AE-2025-0028	2025-05-20	USA	Other	73	FALSE	M	FALSE	Pulmoc
8	AE-2025-0038	2025-03-02	France	Pharmacist	71	FALSE	F	FALSE	Pulmoc
9	AE-2025-0046	2025-06-26	Brazil	Other	54.5	TRUE	M	FALSE	Rheuma
10	AE-2025-0064	2025-07-09	Australia	Physician	57	FALSE	M	FALSE	Glucom
11	AE-2025-0039	2025-03-30	Australia	Consumer	27	FALSE	M	FALSE	Pulmoc
12	AE-2025-0061	2025-07-11	Germany	Consumer	37	FALSE	M	FALSE	Oncoze
13	AE-2025-0081	2025-07-08	India	Consumer	54.5	TRUE	M	FALSE	Neurole
14	AE-2025-0080	2025-03-11	Brazil	Physician	26	FALSE	U	FALSE	Glucom
15	AE-2025-0003	2025-07-13	United Kingdom	Nurse	54.5	TRUE	U	TRUE	Neurole
16	AE-2025-0047	2025-10-30	France	Physician	54.5	TRUE	M	FALSE	Vasculi
17	AE-2025-0050	2025-12-08	Germany	Consumer	54.5	TRUE	U	TRUE	Oncoze
18	AE-2025-0057	2025-02-05	India	Nurse	44	FALSE	F	FALSE	Pul
19	AE-2025-0001	2025-01-13	Canada	Physician	49	FALSE	F	FALSE	Ne
20	AE-2025-0007	2025-05-17	USA	Other	54.5	TRUE	M	FALSE	Vasculi
21	AE-2025-0017	2025-05-22	Japan	Physician	54.5	TRUE	F	FALSE	Cardioz
22	AE-2025-0034	2025-10-12	USA	Consumer	84	FALSE	M	FALSE	Oncoze

6.3 **Data Preparation**

Before loading into the STAGING schema, the following transformations were performed in Dataiku DSS:

- Duplicate REPORT_ID records removed.
- Text values standardized.
- Date fields converted to **YYYY-MM-DD** format.
- Missing values reviewed and handled.
- Data types standardized.
- Data quality validation performed.

These transformations ensured that only validated records entered the staging layer.

7.CURATED Schema

7.1 Overview

The **CURATED** schema represents the business-ready analytical layer of the Snowflake data warehouse. This schema contains the dimensional model, analytical views, aggregation tables, and machine learning datasets required for reporting and AI-driven safety signal detection.

A **Star Schema** was implemented to optimize query performance and simplify analytical reporting.

7.2 **Objectives**

The CURATED schema was designed to:

- Implement a production-ready dimensional model.
- Improve analytical query performance.
- Support business reporting.
- Provide datasets for Snowflake Cortex ML.
- Supply trusted data to Power BI dashboards.

7.3 Star Schema Design

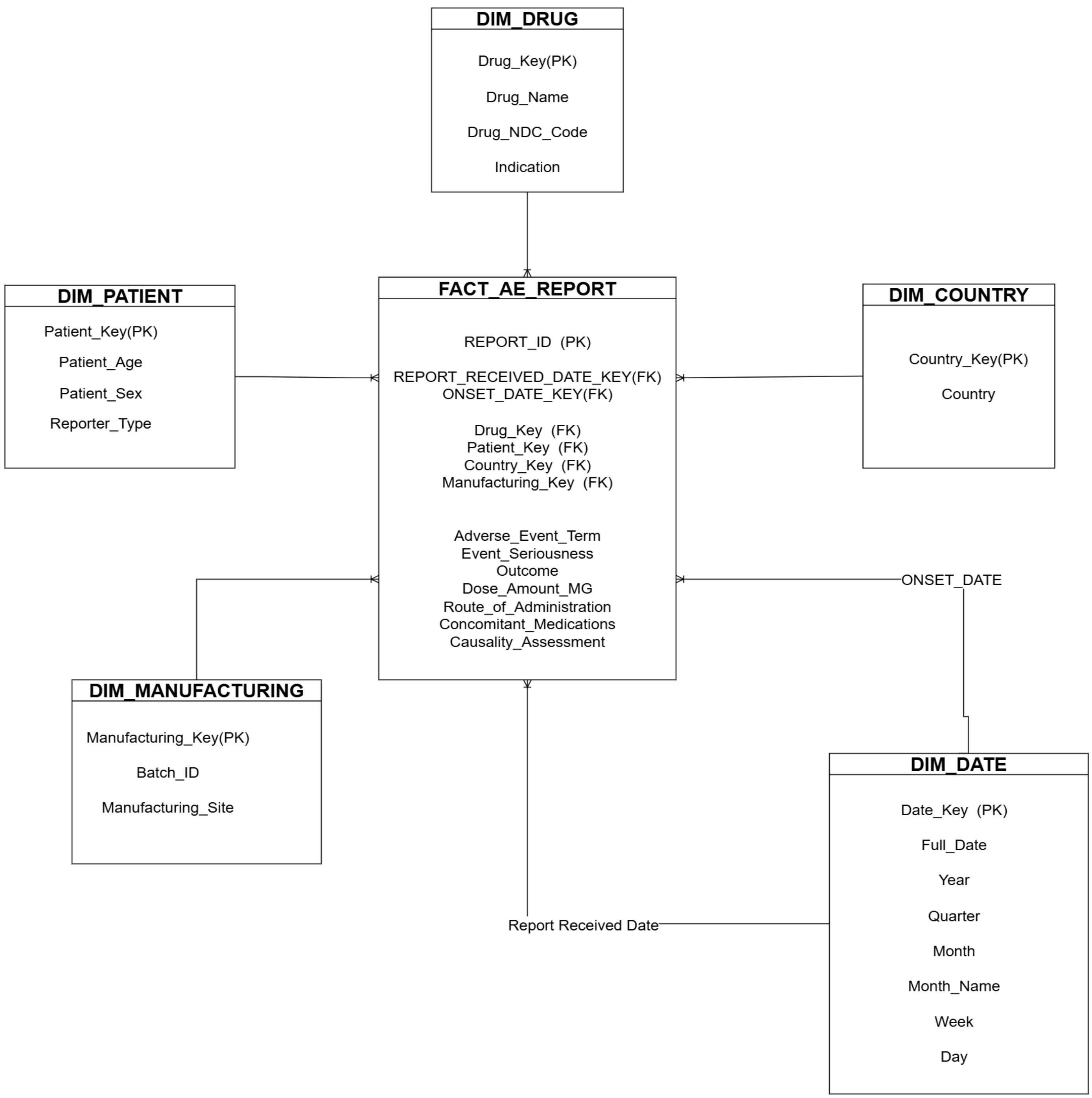
The CURATED schema follows a star schema consisting of one central fact table connected to multiple dimension tables.

FACT TABLE

Table	Purpose
FACT_AE_REPORT	Stores one record for each adverse event report.

Dimension Tables

Dimension	Purpose
DIM_DRUG	Drug information
DIM_PATIENT	Patient demographic information
DIM_COUNTRY	Reporting country information
DIM_MANUFACTURING	Manufacturing batch and site information
DIM_DATE	Calendar attributes for reporting and onset dates



7.4 Surrogate Keys

To improve performance and maintain referential integrity, surrogate keys were generated for all dimension tables.

These surrogate keys replace business values in the fact table and enable efficient joins between fact and dimension tables.

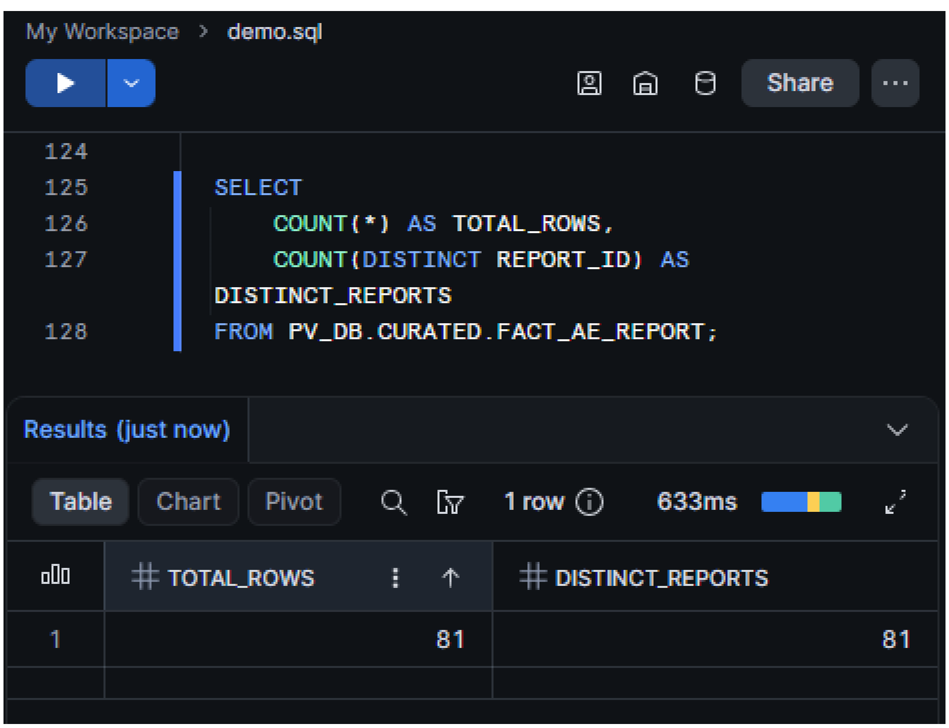
Success Criteria

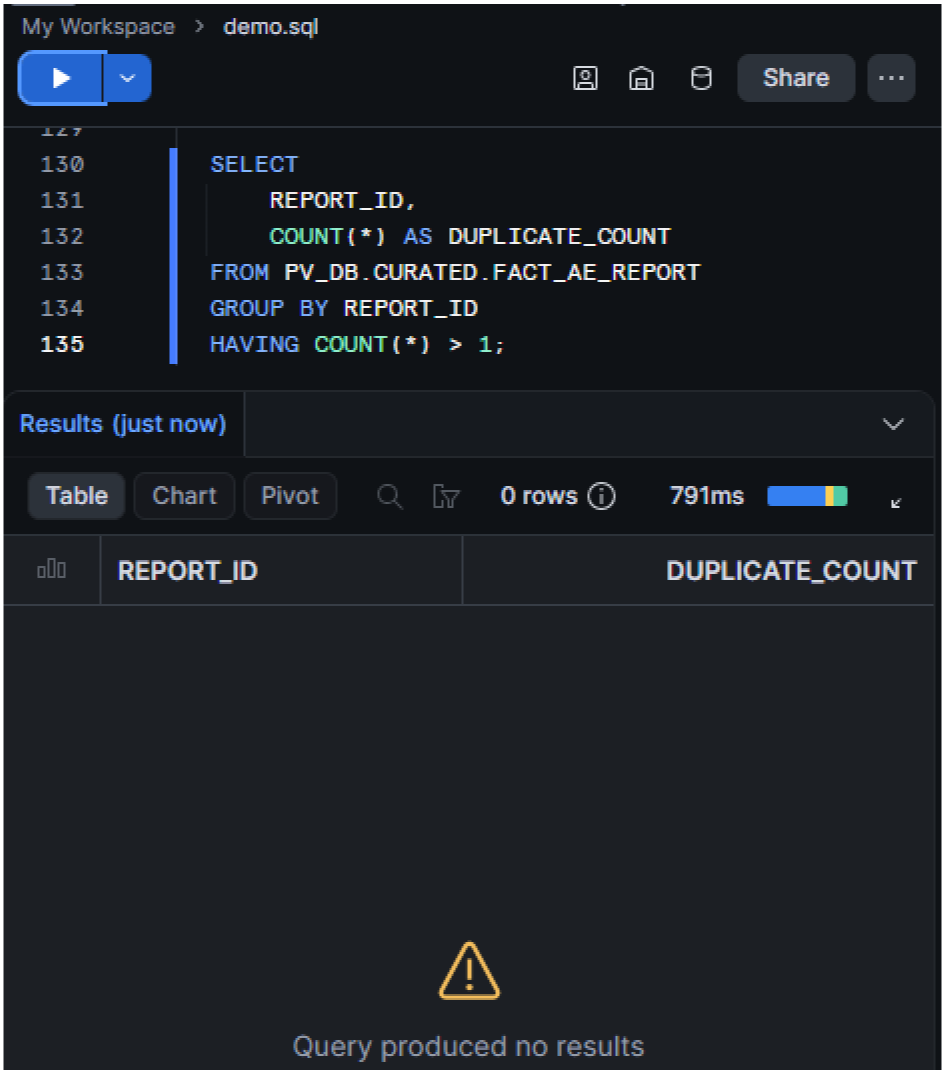
7.5 FACT Table Grain

The grain of the fact table is defined as:

One row represents one adverse event report.

Each record stores surrogate keys that reference the related dimension tables, along with event-specific attributes such as seriousness, outcome, dosage, route of administration, and causality assessment.





7.6 **Referential Integrity Validation**

After loading the fact table, referential integrity checks were performed to verify that every foreign key correctly referenced a corresponding dimension record.

The following relationships were validated:

- Drug Key
- Patient Key
- Country Key
- Manufacturing Key
- Report Received Date Key
- Onset Date Key

All validation queries returned zero orphan records, confirming successful data loading.

```sql
-- Validation 1 : Drug Dimension
---------------------------------------------------------------
SELECT COUNT(*) AS MISSING_DRUG_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_DRUG D
ON F.DRUG_KEY = D.DRUG_KEY
WHERE D.DRUG_KEY IS NULL;

---------------------------------------------------------------
-- Validation 2 : Patient Dimension
---------------------------------------------------------------
SELECT COUNT(*) AS MISSING_PATIENT_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_PATIENT P
ON F.PATIENT_KEY = P.PATIENT_KEY
WHERE P.PATIENT_KEY IS NULL;

---------------------------------------------------------------
-- Validation 3 : Country Dimension
---------------------------------------------------------------
SELECT COUNT(*) AS MISSING_COUNTRY_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_COUNTRY C
ON F.COUNTRY_KEY = C.COUNTRY_KEY
WHERE C.COUNTRY_KEY IS NULL;

---------------------------------------------------------------
-- Validation 4 : Manufacturing Dimension
---------------------------------------------------------------
SELECT COUNT(*) AS MISSING_MANUFACTURING_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_MANUFACTURING M
ON F.MANUFACTURING_KEY = M.MANUFACTURING_KEY
WHERE M.MANUFACTURING_KEY IS NULL;
```

```sql
------------------------------------------------
---------------
-- Validation 5 : Report Received Date
------------------------------------------------
---------------

SELECT COUNT(*) AS INVALID_REPORT_DATE_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_DATE D
ON F.REPORT_RECEIVED_DATE_KEY = D.DATE_KEY
WHERE D.DATE_KEY IS NULL;
```

```sql
-- Validation 6 : Onset Date
------------------------------------------------
---------------
SELECT COUNT(*) AS INVALID_ONSET_DATE_KEYS
FROM FACT_AE_REPORT F
LEFT JOIN DIM_DATE D
ON F.ONSET_DATE_KEY = D.DATE_KEY
WHERE D.DATE_KEY IS NULL;
```

7.7 Analytical Views

To simplify business reporting, analytical views were created within the CURATED schema.

View	Purpose
VW_AE_COUNTS_BY_DRUG_MONTH	Monthly adverse event counts by drug
VW_SERIOUS_EVENT_RATE_BY_COUNTRY	Serious adverse event rate by country

These views provide reusable datasets for reporting and analytical queries.

My Workspace > 09_Analytical_Views.sql

```sql
-- =====================================================================
-- View 1 : Monthly Adverse Event Count by Drug
-- Purpose :
-- Returns monthly adverse event counts for each drug.
-- Used for trend analysis and reporting.
-- =====================================================================

CREATE OR REPLACE VIEW VW_AE_COUNTS_BY_DRUG_MONTH AS
SELECT
    D.DRUG_NAME,
    DT.YEAR,
    DT.MONTH,
    DT.MONTH_NAME,
    COUNT(F.REPORT_ID) AS AE_COUNT
FROM FACT_AE_REPORT F
JOIN DIM_DRUG D
ON F.DRUG_KEY = D.DRUG_KEY
JOIN DIM_DATE DT
ON F.REPORT_RECEIVED_DATE_KEY = DT.DATE_KEY
GROUP BY
    D.DRUG_NAME,
    DT.YEAR,
    DT.MONTH,
    DT.MONTH_NAME
ORDER BY
    D.DRUG_NAME,
    DT.YEAR,
    DT.MONTH;
-- ---------------------------------------------------------------
-- Validation
-- ---------------------------------------------------------------

SELECT *
FROM VW_AE_COUNTS_BY_DRUG_MONTH;
```

Table Chart Pivot 57 rows 824ms

	DRUG_NAME	YEAR	MONTH	MONTH_NAME	AE_COUNT
	Vasculin 19.3% Glucomend 12.3% +7 more	2025 2025	1 12	Jul 14.0% May 14.0% +10 more	1 4
1	Cardiozan	2025	2	Feb	2
2	Cardiozan	2025	3	Mar	1
3	Cardiozan	2025	5	May	1
4	Cardiozan	2025	7	Jul	4
5	Cardiozan	2025	8	Aug	1
6	Cardiozan	2025	11	Nov	1
7	Dermafex	2025	2	Feb	2
8	Dermafex	2025	3	Mar	2
9	Dermafex	2025	5	May	1
10	Dermafex	2025	7	Jul	1
11	Dermafex	2025	11	Nov	1
12	Glucomend	2025	1	Jan	1
13	Glucomend	2025	3	Mar	1
14	Glucomend	2025	5	May	2
15	Glucomend	2025	6	Jun	1
16	Glucomend	2025	7	Jul	3
17	Glucomend	2025	11	Nov	2
18	Glucomend	2025	12	Dec	1
19	Immunova	2025	5	May	1
20	Immunova	2025	6	Jun	1
21	Immunova	2025	7	Jul	1
22	Immunova	2025	9	Sep	1
23	Immunova	2025	11	Nov	1
24	Immunova	2025	12	Dec	1
25	Neurolex	2025	1	Jan	1
26	Neurolex	2025	5	May	1
27	Neurolex	2025	7	Jul	2
28	Neurolex	2025	11	Nov	1
29	Neurolex	2025	12	Dec	2

```sql
CREATE OR REPLACE VIEW VW_SERIOUS_EVENT_RATE_BY_COUNTRY AS
SELECT
    C.COUNTRY,
    COUNT(F.REPORT_ID) AS TOTAL_REPORTS,
    SUM(
        CASE
            WHEN F.EVENT_SERIOUSNESS='Serious'
            THEN 1
            ELSE 0
        END
    ) AS SERIOUS_REPORTS,
    ROUND(
        SUM(
            CASE
                WHEN F.EVENT_SERIOUSNESS='Serious'
                THEN 1
                ELSE 0
            END
        )
        *100.0
        /COUNT(F.REPORT_ID),
        2
    ) AS SERIOUS_RATE_PERCENT
FROM FACT_AE_REPORT F
JOIN DIM_COUNTRY C
ON F.COUNTRY_KEY = C.COUNTRY_KEY
GROUP BY
C.COUNTRY
ORDER BY
SERIOUS_RATE_PERCENT DESC;
---------------------------------------------------------------
-- Validation
---------------------------------------------------------------

SELECT *
FROM VW_SERIOUS_EVENT_RATE_BY_COUNTRY;
```

	COUNTRY	TOTAL_REPORTS	SERIOUS_REPORTS	SERIOUS_RATE_PERCENT
1	Japan	5	5	100.00
2	Australia	10	6	60.00
3	Canada	10	6	60.00
4	Brazil	14	8	57.14
5	United Kingdom	9	5	55.56
6	USA	10	4	40.00
7	India	8	3	37.50
8	Germany	11	3	27.27
9	France	4	0	0.00

8.Data Dictionary

8.1 Overview

A Data Dictionary provides a detailed description of the business tables and columns implemented in the Snowflake CURATED schema. It serves as a reference document for developers, analysts, and business users by defining the meaning, data type, and purpose of each attribute within the dimensional model.

8.2 FACT_AE_REPORT

Table Description

Stores one record for each adverse event report.

Table Grain

One row = One Adverse Event Report

Column	Data Type	Description
REPORT_ID	VARCHAR	Unique adverse event report identifier
REPORT_RECEIVED_DATE_KEY	NUMBER	Foreign key referencing DIM_DATE
ONSET_DATE_KEY	NUMBER	Foreign key referencing DIM_DATE
DRUG_KEY	NUMBER	Foreign key referencing DIM_DRUG
PATIENT_KEY	NUMBER	Foreign key referencing DIM_PATIENT

COUNTRY_KEY	NUMBER	Foreign key referencing DIM_COUNTRY
MANUFACTURING_KEY	NUMBER	Foreign key referencing DIM_MANUFACTURING
ADVERSE_EVENT_TERM	VARCHAR	Reported adverse event
EVENT_SERIOUSNESS	VARCHAR	Serious / Non-Serious classification
OUTCOME	VARCHAR	Outcome of the adverse event
DOSE_AMOUNT_MG	FLOAT	Drug dosage administered
ROUTE_OF_ADMINISTRATION	VARCHAR	Route of drug administration
CONCOMITANT_MEDICATIONS	VARCHAR	Additional medications taken
CAUSALITY_ASSESSMENT	VARCHAR	Initial causality assessment

8.3 DIM_DRUG

Column	Data Type	Description
DRUG_KEY	NUMBER	Surrogate Key
DRUG_NAME	VARCHAR	Name of drug
DRUG_NDC_CODE	VARCHAR	National Drug Code
INDICATION	VARCHAR	Therapeutic indication

8.4 DIM_PATIENT

Column	Description
PATIENT_KEY	Surrogate Key
PATIENT_AGE	Patient age
PATIENT_SEX	Gender
REPORTER_TYPE	Healthcare professional / Consumer

8.5 DIM_COUNTRY

Column	Description
COUNTRY_KEY	Surrogate Key
COUNTRY	Reporting country

8.6 DIM_MANUFACTURING

Column	Description
MANUFACTURING_KEY	Surrogate Key
BATCH_ID	Manufacturing batch
MANUFACTURING_SITE	Manufacturing site

8.7 DIM_DATE

Column	Description
DATE_KEY	Surrogate Key
FULL_DATE	Calendar date
DAY	Day of month
MONTH	Month number
MONTH_NAME	Month name
QUARTER	Quarter
YEAR	Calendar year

9.Snowflake Cortex ML

9.1 Overview

To enable early detection of potential drug safety issues, Snowflake Cortex ML was used to implement an anomaly detection model on aggregated adverse event (AE) reporting data. Rather than analyzing individual adverse event reports, the fact table was aggregated into monthly adverse event counts by drug, creating a time-series dataset suitable for machine learning.

The model was trained using historical adverse event reporting patterns and then evaluated on future observations to identify statistically unusual increases in reporting frequency. The output provides early warning signals that may require further clinical investigation.

9.2 Objectives

The Snowflake Cortex ML implementation was designed to:

- Detect unusual spikes in adverse event reporting.
- Identify statistically significant deviations from historical reporting patterns.
- Support early pharmacovigilance signal detection.
- Generate AI-assisted insights for safety reviewers.
- Demonstrate the integration of machine learning within the Snowflake platform.

9.3 Aggregation Layer

To support time-series machine learning, aggregated tables were created from the fact table.

The primary aggregation implemented was:

AE_COUNTS_BY_DRUG_MONTH and AE_COUNTS_BY_DRUG_WEEK

This table contains:

- Drug Name
- Report Month (Same as Week)
- Adverse Event Count

This aggregated dataset was used as the input for Snowflake Cortex ML anomaly detection.


Table Chart Pivot 57 rows 66ms

	DRUG_NAME	YEAR	MONTH	MONTH_NAME	AE_COUNT
	Vasculin 19.3% Glucomend 12.3% +7 more	2025 2025	1 12	Jul 14.0% May 14.0% +10 more	1 4
1	Cardiozan	2025	2	Feb	2
2	Cardiozan	2025	3	Mar	1
3	Cardiozan	2025	5	May	1
4	Cardiozan	2025	7	Jul	4
5	Cardiozan	2025	8	Aug	1
6	Cardiozan	2025	11	Nov	1
7	Dermafex	2025	2	Feb	2
8	Dermafex	2025	3	Mar	2
9	Dermafex	2025	5	May	1
10	Dermafex	2025	7	Jul	1
11	Dermafex	2025	11	Nov	1
12	Glucomend	2025	1	Jan	1
13	Glucomend	2025	3	Mar	1
14	Glucomend	2025	5	May	2
15	Glucomend	2025	6	Jun	1
16	Glucomend	2025	7	Jul	3
17	Glucomend	2025	11	Nov	2
18	Glucomend	2025	12	Dec	1
19	Immunova	2025	5	May	1
20	Immunova	2025	6	Jun	1
21	Immunova	2025	7	Jul	1
22	Immunova	2025	9	Sep	1
23	Immunova	2025	11	Nov	1
24	Immunova	2025	12	Dec	[illegible]
25	Neurolex	2025	1	Jan	[illegible]
26	Neurolex	2025	5	May	1
27	Neurolex	2025	7	Jul	2
28	Neurolex	2025	11	Nov	1
29	Neurolex	2025	12	Dec	2

9.4 <u>Data Preparation</u>

The anomaly detection model requires time-series data.

The curated fact table was aggregated into monthly and weekly adverse event counts grouped by drug.

The aggregated dataset contains the following attributes:

Column	Description
DRUG_NAME	Name of the reported drug
REPORT_MONTH	Month of adverse event reporting
AE_COUNT	Number of adverse events reported during the month

Same As for Weekly based.

The resulting table created for machine learning was:

AE_COUNTS_BY_DRUG_MONTH_ML

9.5 **<u>Training and Prediction Dataset</u>**

To simulate a real-world forecasting scenario, the dataset was divided into separate training and prediction datasets.

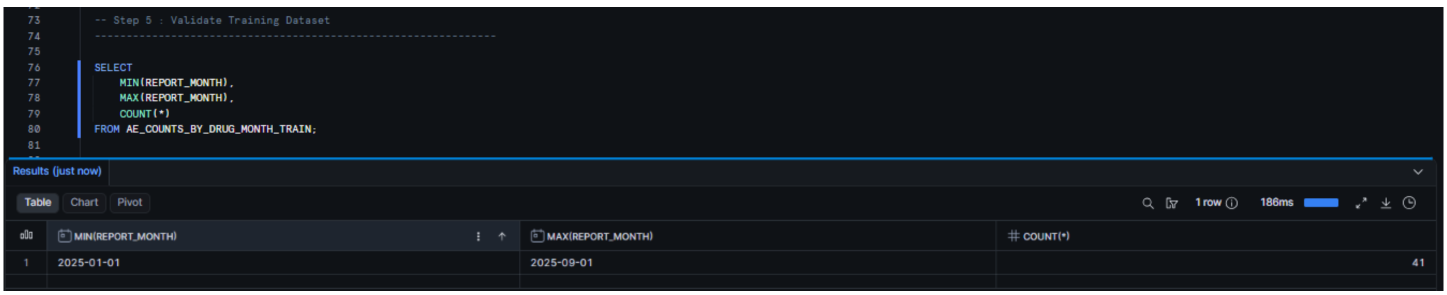
Training Dataset

Table:

AE_COUNTS_BY_DRUG_MONTH_TRAIN

Training Period:

January 2025 – September 2025



```
-- Step 5 : Validate Training Dataset
---------------------------------------------------------------

SELECT
    MIN(REPORT_MONTH),
    MAX(REPORT_MONTH),
    COUNT(*)
FROM AE_COUNTS_BY_DRUG_MONTH_TRAIN;
```

Results (just now)

Table Chart Pivot

1 row 186ms

	MIN(REPORT_MONTH)	MAX(REPORT_MONTH)	COUNT(*)
1	2025-01-01	2025-09-01	41

Purpose:

Historical observations used to train the anomaly detection model.

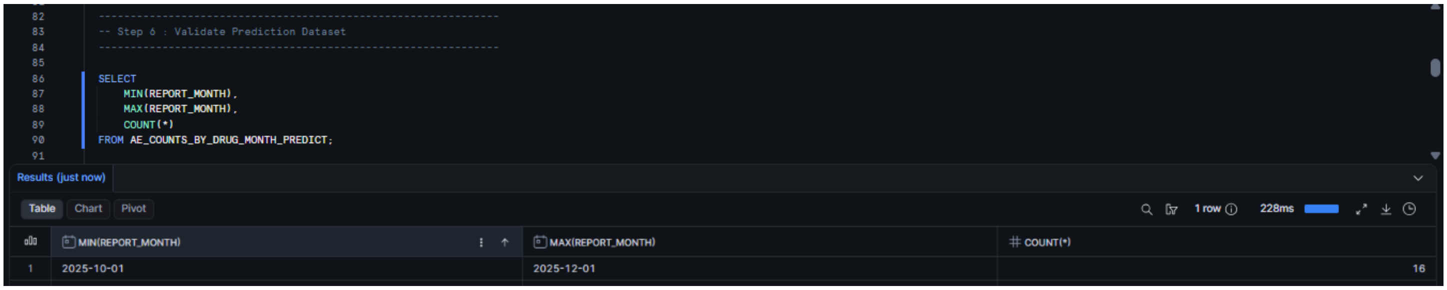
Prediction Dataset

Table:

AE_COUNTS_BY_DRUG_MONTH_PREDICT

Prediction Period:

October 2025 – December 2025



Purpose:

Future observations evaluated by the trained model to identify unusual reporting patterns.

9.6 **Snowflake Cortex ML Model**

Snowflake Cortex ML provides built-in machine learning capabilities without requiring external frameworks.

The following model was implemented:

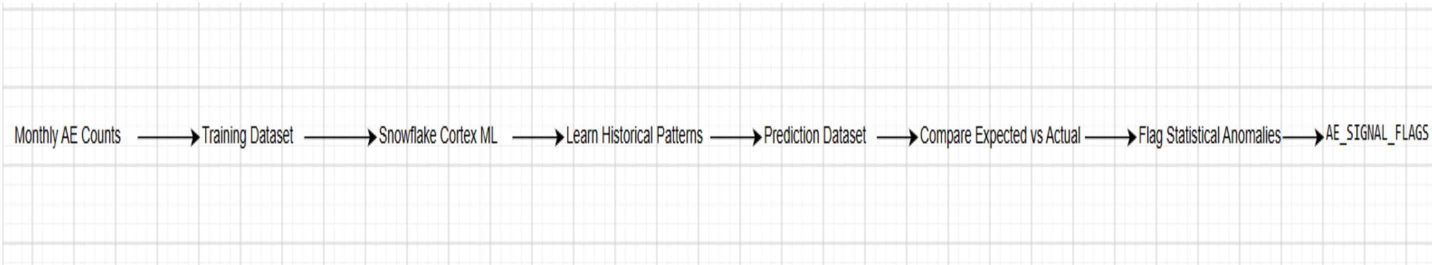
Model	Purpose
SNOWFLAKE.ML.ANOMALY_DETECTION	Detect statistical anomalies in time-series data

The model was configured using:

- **Series Column:** DRUG_NAME
- **Timestamp Column:** REPORT_MONTH
- **Target Column:** AE_COUNT

This configuration allows Snowflake to independently learn historical reporting patterns for each drug.

9.7 **Model Execution**



Success Criteria

9.8 **<u>Prediction Interval(End to End ML)</u>**

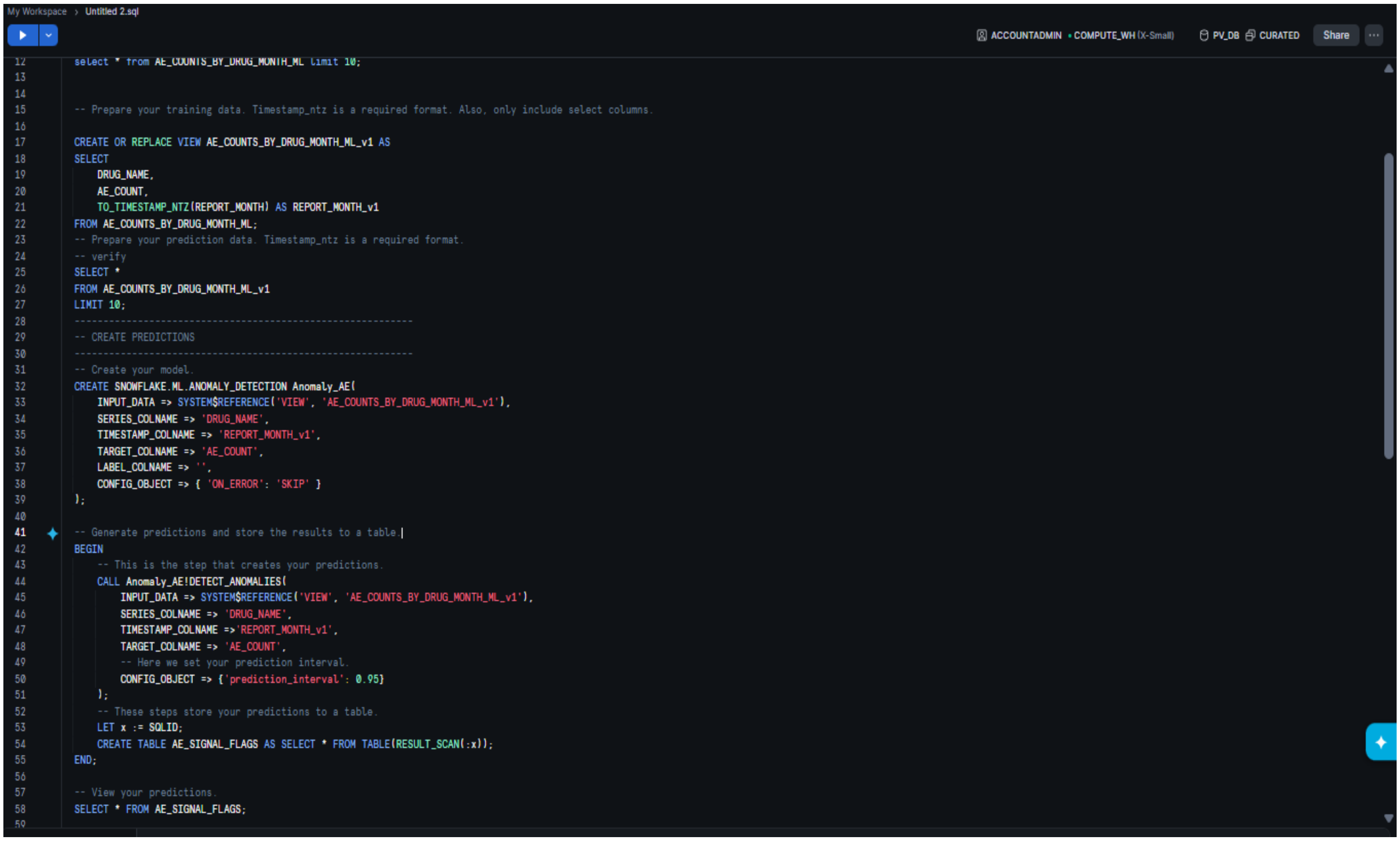
The anomaly detection model was executed with a prediction interval of:

0.95 (95%)

A prediction interval of 95% indicates that the model expects approximately 95% of normal observations to fall within the calculated forecast range.

Observations falling outside this expected range are flagged as potential statistical anomalies.

This threshold provides a balance between sensitivity and false positive detection.



9.9 **AI Output**

The anomaly detection results were stored in:

CURATED.AE_SIGNAL_FLAGS

The output includes:

Column	**Description**
SERIES	Drug Name
TS	Reporting Month
Y	Actual Adverse Event Count
FORECAST	Expected Adverse Event Count
LOWER_BOUND	Lower prediction interval
UPPER_BOUND	Upper prediction interval
IS_ANOMALY	Statistical anomaly flag

Only 1 Anomaly Detected for drug per month (ONCOZEL).

2 Anomaly Detected for drug per week (ONCOZEL,VASCULIN)

	SERIES	TS	Y	FORECAST	LOWER_BOUND	UPPER_BOUND	IS_ANOMALY	PERCENTILE	DISTANCE
1	"Cardiozan"	2025-11-01 00:00:00.000	1	2.5	0.3086936486	4.691306351	FALSE	0.08985624744	-1.341640786
2	"Rheumatrix"	2025-10-01 00:00:00.000	1	2	-0.7718076487	4.771807649	FALSE	0.2397500611	-0.7071067812
3	"Immunova"	2025-11-01 00:00:00.000	1	1	1	1	FALSE	null	null
4	"Immunova"	2025-12-01 00:00:00.000	1	1	1	1	FALSE	null	null
5	"Glucomend"	2025-11-01 00:00:00.000	2	2	0.3996961079	3.600303892	FALSE	0.5	0
6	"Glucomend"	2025-12-01 00:00:00.000	1	2	0.3996961079	3.600303892	FALSE	0.110335681	-1.224744871
7	"Oncozel"	2025-10-01 00:00:00.000	1	1	1	1	FALSE	null	null
8	"Oncozel"	2025-12-01 00:00:00.000	2	1	1	1	TRUE	1	inf
9	"Pulmocare-XR"	2025-10-01 00:00:00.000	1	1	-0.6003038921	2.600303892	FALSE	0.5	0
10	"Pulmocare-XR"	2025-11-01 00:00:00.000	1	1	-0.6003038921	2.600303892	FALSE	0.5	0
11	"Neurolex"	2025-11-01 00:00:00.000	1	2	0.4145760494	3.585423951	FALSE	0.1081847441	-1.236239672
12	"Neurolex"	2025-12-01 00:00:00.000	2	2	0.3996961079	3.600303892	FALSE	0.5	0
13	"Vasculin"	2025-10-01 00:00:00.000	1	1	-0.2830976169	2.283097617	FALSE	0.5	0
14	"Vasculin"	2025-11-01 00:00:00.000	2	1	-0.8145740517	2.814574052	FALSE	0.8599563946	1.08012345
15	"Vasculin"	2025-12-01 00:00:00.000	1	1	-1.222390264	3.222390264	FALSE	0.5	0
16	"Dermafex"	2025-11-01 00:00:00.000	1	1	-0.9599639845	2.959963985	FALSE	0.5	0

	SERIES	TS	Y	FORECAST	LOWER_BOUND	UPPER_BOUND	IS_ANOMALY	PERCENTILE	DISTANCE
1	"Pulmocare-XR"	2025-10-27 00:00:00.000	1	1	0.1840351964	1.815964804	FALSE	0.5	0
2	"Pulmocare-XR"	2025-11-24 00:00:00.000	1	1	[illegible]	92	FALSE	0.5	0
3	"Oncozel"	2025-10-06 00:00:00.000	1	1	1	1	FALSE	null	null
4	"Oncozel"	2025-12-08 00:00:00.000	2	1	1	1	TRUE	1	inf
5	"Dermafex"	2025-11-10 00:00:00.000	1	1	0.3070480878	1.692951912	FALSE	0.5	0
6	"Cardiozan"	2025-11-17 00:00:00.000	1	1.5	0.8802049677	2.119795032	FALSE	0.056923149	-1.58113883
7	"Rheumatrix"	2025-10-27 00:00:00.000	1	2	0.6140961757	3.385903824	FALSE	0.07864960353	-1.414213562
8	"Immunova"	2025-11-24 00:00:00.000	1	1	1	1	FALSE	null	null
9	"Immunova"	2025-12-01 00:00:00.000	1	1	1	1	FALSE	null	null
10	"Neurolex"	2025-11-24 00:00:00.000	1	1.5	0.5200180077	2.479981992	FALSE	0.1586552539	-1
11	"Neurolex"	2025-12-08 00:00:00.000	1	1.5	0.5200180077	2.479981992	FALSE	0.1586552539	-1
12	"Neurolex"	2025-12-29 00:00:00.000	1	1.5	0.5200180077	2.479981992	FALSE	0.1586552539	-1
13	"Vasculin"	2025-10-27 00:00:00.000	1	1	0.599924027	1.400075973	FALSE	0.5	0
14	"Vasculin"	2025-11-03 00:00:00.000	2	1	0.599924027	1.400075973	TRUE	0.9999995183	4.898979486
15	"Vasculin"	2025-12-01 00:00:00.000	1	1	0.599924027	1.400075973	FALSE	0.5	0
16	"Glucomend"	2025-11-10 00:00:00.000	1	1.5	0.7052187828	2.294781217	FALSE	0.1087839504	-1.233021077
17	"Glucomend"	2025-11-17 00:00:00.000	1	1.5	0.6998480539	2.300151946	FALSE	0.110335681	-1.224744871
18	"Glucomend"	2025-12-22 00:00:00.000	1	1.5	0.6998480539	2.300151946	FALSE	0.110335681	-1.224744871

9.10 **Statistical Signal vs Clinical Causality**

Statistical Signal

A statistical signal is an unusual increase or decrease in adverse event reporting detected by the machine learning model based on historical data patterns.

Example:

Snowflake Cortex ML detected an unusually high number of adverse event reports for a specific drug during October 2025 compared to historical reporting behavior.

Clinical Causality

Clinical causality determines whether the reported adverse event was actually caused by the drug.

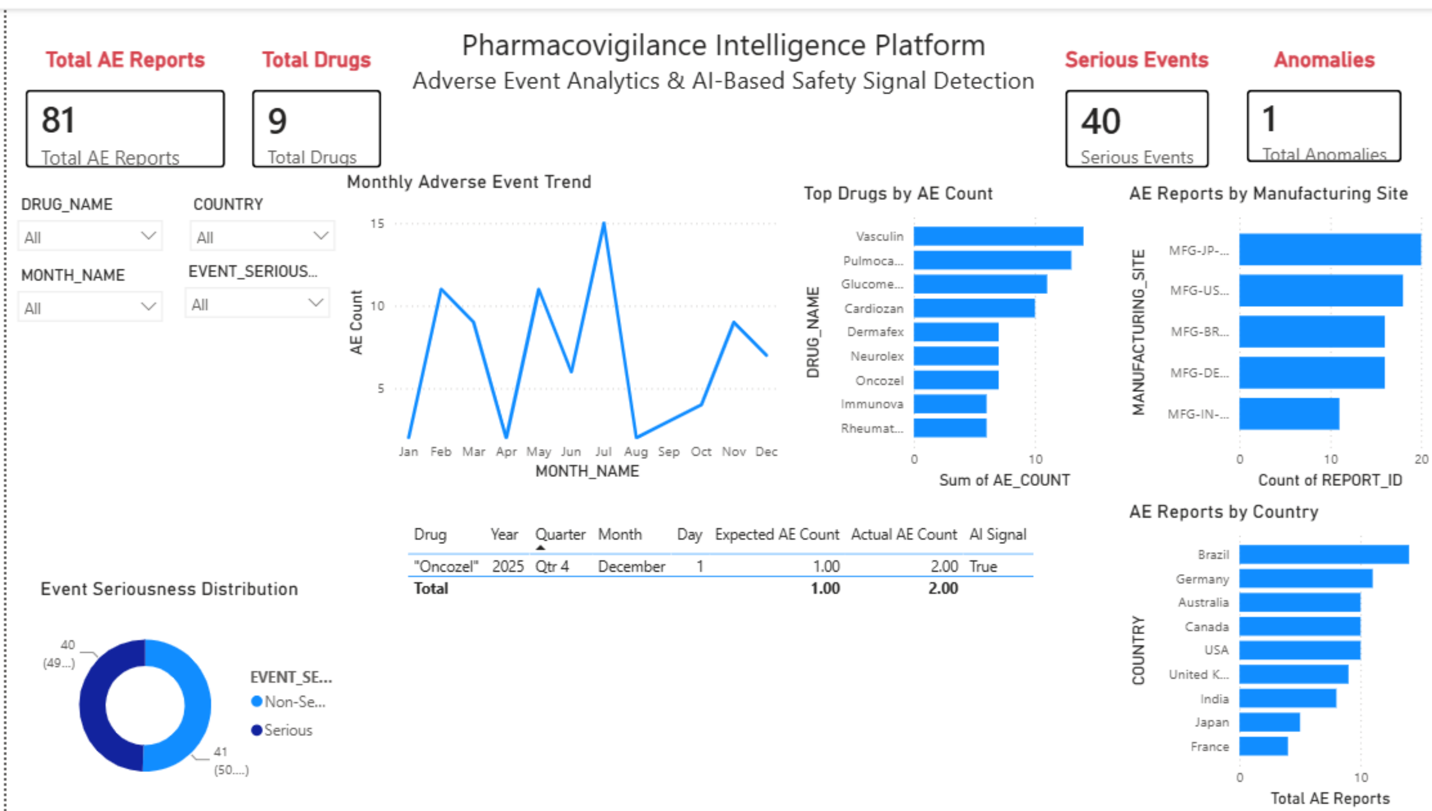
This requires:

- Clinical assessment
- Medical review
- Patient history
- Laboratory evidence
- Regulatory investigation

The machine learning model **does not establish clinical causality**; it only highlights records that warrant further investigation.

10.Dashboard

Drug Per Month



Drug Per Week

Pharmacovigilance Intelligence Platform

Adverse Event Analytics & AI-Based Safety Signal Detection

Total AE Reports

81
Total AE Reports

Total Drugs

9
Total Drugs

Serious Events

40
Serious Events

Anomalies

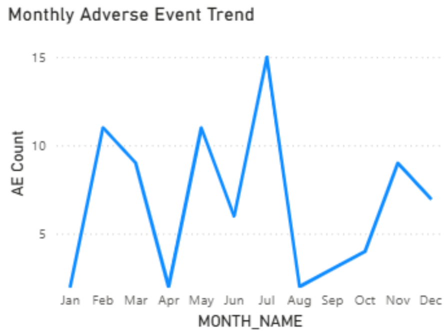
2
Total Anomali...

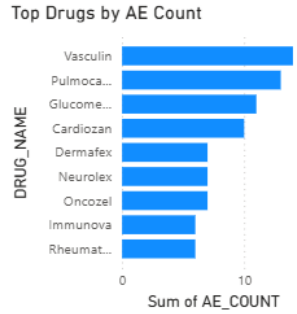
DRUG_NAME: All

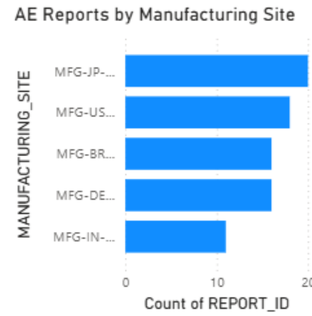
COUNTRY: All

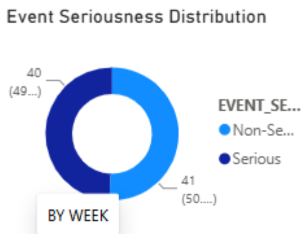
MONTH_NAME: All

EVENT_SERIOU...: All









Drug	Year	Quarter	Month	Day	Expected AE Count	Actual AE Count	AI Signal
"Oncozel"	2025	Qtr 4	December	8	1.00	2.00	True
"Vasculin"	2025	Qtr 4	November	3	1.00	2.00	True
Total					**2.00**	**4.00**	